Copernicus Global Land Operations
"Vegetation and Energy"
"CGLOPS-1"

Framework Service Contract N° –945120 – IPR – 2023

EVALUATION REPORT OF OLCI AND SLSTR
CLOUD, CLOUD SHADOW AND SNOW DETECTION

Issue I2.00

Organization name of lead contractor for this deliverable: VITO

Book Captain: Carolien Toté (VITO)

Contributing Authors: Marc Padilla (Complutig)

Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
RE	Restricted to a group specified by the consortium (including the Commission Services)	
CO	Confidential, only for members of the consortium (including the Commission Services)	

© CGLOPS Lot-1 consortium

Document Release Sheet

Book captain:	Carolien Toté	Sign Carolien Toté	Date 09.12.2024
Approval:	Roselyne Lacaze	Sign Roselyne Lacaze	Date 12.12.2024
Endorsement:	Nadine Gobron	Sign	Date
Distribution:	Public		

© CGLOPS Lot-1 consortium

Change Record

Date	Page(s)	Description of Change	Release
28.09.2019	All	First draft	D1.00
20.05.2020	All	Update after implementation of IdePix upgrade	I1.00
11.02.2022	13, 15, 16, 17 45 - 48 55, 60, 61	Update after external review	I1.10
09.12.2024	All	Update for TOC reflectance V2.3	I2.00

© CGLOPS Lot-1 consortium

TABLE OF CONTENTS

Executive Summary *10*

1 Background of the document *11*

1.1 Scope and Objectives **11**

1.2 Content of the document **11**

1.3 Related documents **11**

1.3.1 Applicable documents 11

1.3.2 Input 11

1.3.3 External documents 12

2 Description of Sentinel-3 Cloud, Cloud shadow and Snow Masking *13*

2.1 Introduction **13**

2.2 Sentinel-3 OLCI IdePix **13**

2.3 Sentinel-3 SLSTR: Level 1B quality flag **15**

3 Evaluation Methodology *16*

3.1 Sentinel-3 OLCI and SLSTR TOC reflectance **16**

3.2 Evaluation for the retrieval of vegetation variables **16**

3.3 Evaluation for Burned Areas detection **17**

4 Results *19*

4.1 Evaluation for the retrieval of vegetation variables **19**

4.1.1 Detection over dark surfaces 19

4.1.2 Detection over bright surfaces 29

4.1.3 Semi-transparent clouds 37

4.1.4 Snow detection 45

4.2 Evaluation for Burned Areas detection **49**

5 Conclusions *58*

6 References *59*

© CGLOPS Lot-1 consortium

List of Figures

Figure 1: IdePix flag coding in S3 TOC Reflectance products 14
Figure 2: OLCI IdePix cloud validation results over land [OPT-MPC, 2024] 15
Figure 3: SLSTR 'cloud_an' flag codings 15
Figure 4: Grid tiles on world map. Red squares indicate tiles under consideration. 17
Figure 5: Color coding to visualize the pixel identification flags in the following figures 19
Figure 6: Product cgl_TOC_20180701072059_X22Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 19
Figure 7: Product cgl_TOC_20180701073829_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 20
Figure 8: Product cgl_TOC_20180701103957_X18Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 20
Figure 9: Product cgl_TOC_20180701104257_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 21
Figure 10: Product cgl_TOC_20180701104257_X18Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 21
Figure 11: Product cgl_TOC_20180701104527_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 22
Figure 12: Product cgl_TOC_20180701141956_X10Y08_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 22
Figure 13: Product cgl_TOC_20181101010915_X30Y10_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 23
Figure 14: Product cgl_TOC_20181101074412_X20Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 23
Figure 15: Product cgl_TOC_20181101074712_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 24
Figure 16: Product cgl_TOC_20181101142809_X10Y08_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 24
Figure 17: Product cgl_TOC_20190301001532_X32Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 25
Figure 18: Product cgl_TOC_20190301072957_X22Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 25
Figure 19: Product cgl_TOC_20190301083128_X20Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 26
Figure 20: Product cgl_TOC_20190301151825_X10Y08_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 26
Figure 21: Product cgl_TOC_20190301141653_X10Y08_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 27
Figure 22: Product cgl_TOC_20190301152125_X10Y08_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 27
Figure 23: Product cgl_TOC_20190601015327_X29Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 28
Figure 24: Product cgl_TOC_20190601023003_X29Y03_S3A_v2.3.4 (RGB Oa17, Oa06, Oa03) 28
Figure 25: Product cgl_TOC_20190601074801_X20Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 29
Figure 26: Product cgl_TOC_20180701005432_X30Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 29
Figure 27: Product cgl_TOC_20180701005432_X30Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 30
Figure 28: Product cgl_TOC_20180701005732_X30Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 30
Figure 29: Product cgl_TOC_20181101092211_X19Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 31
Figure 30: Product cgl_TOC_20190301065029_X22Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 31

Figure 31: Product cgl_TOC_20190301065329_X22Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 32
Figure 32: Product cgl_TOC_20190301065329_X22Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 32
Figure 33: Product cgl_TOC_20190301083128_X19Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 33
Figure 34: Product cgl_TOC_20190301091056_X19Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 33
Figure 35: Product cgl_TOC_20190301234921_X32Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 34
Figure 36: Product cgl_TOC_20190301235221_X32Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 34
Figure 37: Product cgl_TOC_20190601003027_X32Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 34
Figure 38: Product cgl_TOC_20190601003327_X30Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 35
Figure 39: Product cgl_TOC_20190601011004_X30Y10_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 35
Figure 40: Product cgl_TOC_20190601065624_X22Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 36
Figure 41: Product cgl_TOC_20190601074501_X20Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 36
Figure 42: Product cgl_TOC_20190601084624_X19Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 37
Figure 43: Product cgl_TOC_20180701072029_X22Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 37
Figure 44: Product cgl_TOC_20180701021731_X29Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 38
Figure 45: Product cgl_TOC_20180701073559_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 38
Figure 46: Product cgl_TOC_20181101010615_X30Y10_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 39
Figure 47: Product cgl_TOC_20181101022614_X29Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 39
Figure 48: Product cgl_TOC_20181101073212_X22Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 40
Figure 49: Product cgl_TOC_20181101075012_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 40
Figure 50: Product cgl_TOC_20181101092511_X19Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 41
Figure 51: Product cgl_TOC_20181101105711_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 41
Figure 52: Product cgl_TOC_20190301005500_X30Y10_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 42
Figure 53: Product cgl_TOC_20190301073857_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 42
Figure 54: Product cgl_TOC_20190301083428_X19Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 43
Figure 55: Product cgl_TOC_20190301100627_X17Y04_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 43
Figure 56: Product cgl_TOC_20190301104555_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 44
Figure 57: Product cgl_TOC_20190601065324_X22Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 44
Figure 58: Product cgl_TOC_20181101022914_X29Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 45
Figure 59: Product cgl_TOC_20181101072912_X22Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 45
Figure 60: Product cgl_TOC_20181101105411_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 46
Figure 61: Product cgl_TOC_20190301021759_X29Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 46

© CGLOPS Lot-1 consortium

Figure 62: Product cgl_TOC_20190301031630_X29Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 47

Figure 63: Product cgl_TOC_20190301104255_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03) 47

Figure 64: Product cgl_TOC_20190601033126_X29Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03) 48

Figure 65: Product cgl_TOC_20190404143538_X10Y08_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows large over-estimation of cloud by the “thin_cirrus” from the ‘cloud_an’ layer (red circle). 50

Figure 66: Product cgl_TOC_20180804075137_X20Y07_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows an under-detection of clouds (center-right of the image) by both quality layers. Cloud shadows are mostly missed. 51

Figure 67: Product cgl_TOC_20190113023640_X29Y03_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows good detection of clouds on the ‘pixel_classif_flags’ layer, although with quite large under-estimations of cloud shadow at the center of the image and underestimations of snow (snow cover observed across nearby dates). The ‘cloud_an’ layer omits some of the clouds. 52

Figure 68: Product cgl_TOC_20180705091013_X19Y07_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows how cloud shadows are mostly missed by the ‘pixel_classif_flags’. Some clouds are not detected by any flag, particularly in ‘cloud_an’. 53

Figure 69: Product cgl_TOC_20190622235859_X32Y10_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows how some clouds are not detected by any flag, particularly in ‘cloud_an’. 54

Figure 70: Product cgl_TOC_20190421145902_X10Y08_S3B_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows large over-detection of thin cirrus on the ‘cloud_an’ layer (center-left part of the image). 55

Figure 71: Product cgl_TOC_20190107075436_X22Y03_S3B_v2.3.4.nc (RGB: S1, Oa17, Oa06). This example shows a large under-detection of snow (snow is consistently observed across nearby dates). 56

Figure 72: Product cgl_TOC_20181104012842_X30Y10_S3A_v2.3.4.nc (RGB: S1, Oa17, Oa06). This is an example of clearly excessive thin cirrus detection by the ‘cloud an’ layer. 57

List of Acronyms

AC	Atmospheric Correction
AD	Applicable Document
ATBD	Algorithm Theoretical Basis Document
BA	Burnt Areas
CCI	Climate Change Initiative
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
DMP	Dry Matter Productivity
ESA	European Space Agency
FAPAR	Fraction of Absorbed Photosynthetically Active Radiation
FCOVER	Fraction of green vegetation cover
IdePix	Pixel Identification and classification
L1B	Level 1B
LAI	Leaf Area Index
LC	Land Cover
LSP	Land Surface Phenology
NDVI	Normalized Difference Vegetation Index
NPP	Net Primary Production
OLCI	Ocean and Land Color Instrument
OPT-MPC	Optical Mission Performance Cloud
PQAD	Product Quality Assurance Document
PUM	Product User Manual
QAR	Quality Assessment Report
RGB	Red Green Blue
S3	Sentinel-3
S3VT	Sentinel-3 Validation Team
SLSTR	Sea and Land Surface Temperature Radiometer
SNAP	SeNtinel Application Platform
SWIR	Short Wave Infrared
TIR	Thermal Infrared
TOA	Top-Of-Atmosphere
TOC	Top-Of-Canopy
V	Version
VNIR	Visible and Infrared

© CGLOPS Lot-1 consortium

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters are to take place in a timely manner and are complemented by the constitution of long-term time series.

CLMS portfolio contains a number of global "Vegetation" biophysical products based upon the data acquired by the Ocean and Land Colour Instrument (OLCI) and the Sea and Land Surface Temperature Radiometer (SLSTR) onboard the Sentinel-3 (S3) platform. All various biophysical algorithms ingest the same Top-Of-Canopy (TOC) Reflectances derived from the Level 1B Top-Of-Atmosphere (TOA) data by a harmonized pre-processing chain. One key element of the pre-processing is the identification and the masking of the clouds, the cloud shadows and the snowy pixels.

This report evaluates the performance of the cloud, cloud shadow and snow masking in Sentinel-3 OLCI and SLSTR TOC reflectance Collection 300m V2.3 for both the retrieval of vegetation variables and for burned areas (BA) detection. The masking is performed on OLCI data by the IdePix processor (v9.0.6), along with other quality flags provided by the Level 1B OLCI and SLSTR data. The document describes the masking algorithms, the quality assessment approach and the results of the analysis.

From this evaluation, we conclude that the IdePix cloud mask and cloud shadow mask generally perform well. IdePix however tends to omit thin or semi-transparent clouds and cloud edges. Compared to previous IdePix versions, the snow detection is improved but clear sky snow pixels are still often incorrectly identified as clouds. In some cases, but less frequently compared to previous versions, IdePix over-detects clouds over bright surfaces, such as bare areas and water shorelines.

Regarding the SLSTR cloud flag, this layer performs well to mask clouds but there is over-detection of the thin cirrus mask. For BA detection, the thin cirrus mask is not used. Since the SLSTR 'cloud_an' layer does not contain cloud shadow identification, IdePix is needed to detect cloud shadows in SLSTR images. No residual cloud shadows in the SLSTR S5 data channel (centred in 1613 nm) is observed.

© CGLOPS Lot-1 consortium

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The scope of this document is to provide the results of the evaluation of cloud detection performed, in the Copernicus Global Land Operations (CGLOPS) pre-processing chain, on Sentinel-3 sensors data. Cloud detection is currently performed on Sentinel-3 OLCI and SLSTR data separately, using the IdePix processor (developed by the Sentinel-3 Mission Performance Cloud) for OLCI and through the Level 1B 'cloud_an' layer for SLSTR.

The objective of the evaluation is to clearly identify the limitations of the existing cloud/shadow/snow masking.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 describes the Sentinel-3 cloud, cloud shadow and snow masking algorithms, and the expected performance
- Chapter 3 describes the evaluation methodology
- Chapter 4 reports the results of the analysis
- Chapter 5 summarizes the conclusions

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document for Vegetation and Energy products of the Copernicus Global Land Service

© CGLOPS Lot-1 consortium

CGLOPS1_ATBD_S3-AC-V1.1	Algorithm Theoretical Basis Document – Atmospheric Correction for Sentinel-3 OLCI and SLSTR products Version 1.1
CGLPS1_PUM_TOC-Reflectance300m-V2.3	Product User Manual of the Sentinel-3 Top of Canopy (TOC) Reflectance Collection 300m Version 2.3
CGLOPS1_QAR_TOC-reflectance300m-V2.3	Quality Assessment report of the Sentinel-3 Top of Canopy (TOC) Reflectance Collection 300m Version 2.3

These documents are available at: https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Product Data Format Specification - SLSTR Level1 Products, I2.10, 15 October 2021, available at https://sentiwiki.copernicus.eu/web/document-library#DocumentLibrary-SLSTRLibrary-S3-OLCI-PDS

SLSTR ATBD for Level 1 observables – S3-TN-RAL-SL-032, I8.1, 1 April 2021, available at https://sentiwiki.copernicus.eu/web/document-library#DocumentLibrary-SLSTRLibrary-S3-OLCI-PDS

Sentinel-3 OLCI IdePix Algorithm Theoretical Basis Document, V1.0, Wevers et al., 4 May 2022, available at https://zenodo.org/record/6517333

Land Cover CCI ATBD Version2. CCI-LC-ATBD I2.3, 28 November 2013, available at https://climate.esa.int/en/projects/land-cover/key-documents/

OPT-MPC, 2024. Sentinel-3 optical Annual Performance Report-Year 2023. Available at https://sentiwiki.copernicus.eu/__attachments/1681931/OMPC.ACR.APR.004 - S3 Optical Annual Performance Report 2023 - 1.1.pdf

2 DESCRIPTION OF SENTINEL-3 CLOUD, CLOUD SHADOW AND SNOW MASKING

2.1 INTRODUCTION

In the CGLOPS pre-processing chain, the cloud, cloud shadow and snow masking is based on separate schemes for OLCI (using IdePix) and SLSTR (using Level 1B quality data flag). Both schemes are described in the following sections.

A so-called SYNERGY cloud screening, using the combined information in OLCI and SLSTR and considering the different geometries and different spectral ranges (visible and near infrared (VNIR), short-wave infrared (SWIR) and thermal infrared (TIR)) with different optical thickness due to clouds is still in development and currently not available (Wevers et al., 2019).

2.2 SENTINEL-3 OLCI IDEPIX

The IdePix processor provides a pixel classification into properties such as clear/cloudy, land/water, snow, ice etc. The processing options/parameters as well as the underlying classification algorithms are instrument-dependent. The IdePix classification algorithm for Sentinel-3 OLCI is based on a neural network approach. A common neural network is used for both land and water pixels. As input for the neural network, the square roots of the OLCI TOA reflectances (obtained from an internal radiance-to-reflectance conversion) in all 21 OLCI bands are used. As output, the neural network finally provides, per pixel, the properties ‘cloud sure’, ‘cloud ambiguous’, ‘cloud’ (which means ‘sure’ or ‘ambiguous’), ‘snow/ice’ and ‘mountain shadow’. Note that the ‘mountain shadow’ flag is not considered for CGLOPS downstream biophysical processing; hence the quality of this flag is not analysed.

For pixels identified as cloudy, a ‘cloud_buffer’ of 2 pixels wide is set to obtain a more conservative cloud mask. This is done in a post-processing step after the cloud classification has been applied on the full image. The cloud buffer procedure is described in more detail in the [Land Cover CCI ATBD Version2]. The ‘land’ and ‘coastline’ pixels are identified from the land flag included in the quality flags which come with the L1B OLCI product.

The IdePix neural network structure includes detection of clouds over snow, cloud shadow detection (Wevers et al., 2019), and also the ‘bright’, ‘white’ and ‘cloud_shadow’ pixel properties are provided.

The version of IdePix implemented in the CGLOPS pre-processing chain V2.3 is Idepix version 9.0.6[1], with improved discrimination of cloud and snow or lake ice, compared to previous versions.

The IdePix classification result is written into the TOC Reflectance product as a flag band named 'pixel_classif_flags'. This band holds integer values which follow the flag coding shown in Figure 1.

[1] pixel_classif_flags

Name	Value	Type
IDEPIX_INVALID	1	uint32
IDEPIX_CLOUD	2	uint32
IDEPIX_CLOUD_AMBIGUOUS	4	uint32
IDEPIX_CLOUD_SURE	8	uint32
IDEPIX_CLOUD_BUFFER	16	uint32
IDEPIX_CLOUD_SHADOW	32	uint32
IDEPIX_SNOW_ICE	64	uint32
IDEPIX_BRIGHT	128	uint32
IDEPIX_WHITE	256	uint32
IDEPIX_COASTLINE	512	uint32
IDEPIX_LAND	1024	uint32
IDEPIX_MOUNTAIN_SHADOW	2048	uint32

Figure 1: IdePix flag coding in S3 TOC Reflectance products

Based on a previous version of IdePix (v2.16), according to Wevers et al. (2019), user accuracy levels and overall accuracy (86%) of cloud identification were on a satisfying level although omission and commission errors showed that there was a tendency of commissioning bright clear observations as cloud. The new IdePix v9.0.6 is performing better in classifying snow, and reaches an overall accuracy of about 90% over land surfaces (Figure 2).

[1] https://github.com/bcdev/snap-idepix/releases/tag/v9.0.6

© CGLOPS Lot-1 consortium

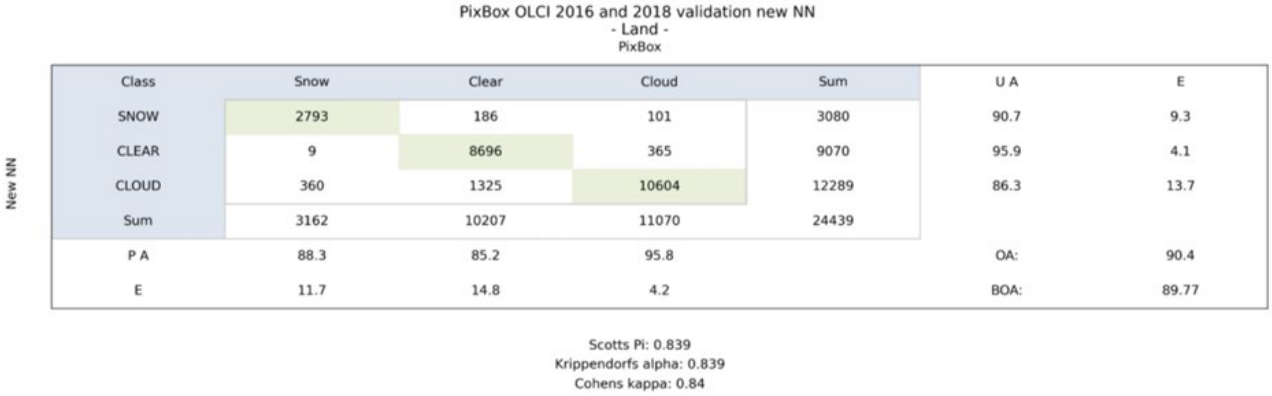

Figure 2: OLCI IdePix cloud validation results over land [OPT-MPC, 2024]

2.3 SENTINEL-3 SLSTR: LEVEL 1B QUALITY FLAG

The SLSTR Level1B data contains the output of a Basic Cloud Identification ('cloud_an'), based on a collection of separate cloud tests. The clouds are identified by using test over the brightness temperature data in the thermal channels and the reflectance data in the short-wave infra-red and visible channels [SLSTR ATBD for Level 1 observables; Section 7.4.3].

The 'cloud_an' flag coding is shown in Figure 3. No accuracy information of the basic cloud tests over land is available.

[1] cloud_an

Name	Value	Type	Description
visible	1	uint32	
1_37_threshold	2	uint32	
1_6_small_histogram	4	uint32	
1_6_large_histogram	8	uint32	
2_25_small_histogram	16	uint32	
2_25_large_histogram	32	uint32	
11_spatial_coherence	64	uint32	
gross_cloud	128	uint32	
thin_cirrus	256	uint32	
medium_high	512	uint32	
fog_low_stratus	1024	uint32	
11_12_view_difference	2048	uint32	
3_7_11_view_difference	4096	uint32	
thermal_histogram	8192	uint32	
spare_1	16384	uint32	
spare_2	32768	uint32	

Figure 3: SLSTR 'cloud_an' flag codings

3 EVALUATION METHODOLOGY

3.1 SENTINEL-3 OLCI AND SLSTR TOC REFLECTANCE

The OLCI and SLSTR TOC reflectances are used as input for the calculation of the Fraction of photosynthetically active radiation absorbed by the vegetation (FAPAR), the lead area index (LAI), the fraction of green vegetation cover (FCOVER), the dry matter productivity (DMP), the net primary production (NPP), the normalized difference vegetation index (NDVI), the land surface phenology (LSP) and for the burnt area detection. The qualitative evaluation of the cloud, cloud shadow and snow detection is done through visualization of a large number of TOC reflectance tiles.

The following naming standard is used:

cgl_TOC_<TIME>_<AREA>_<SATELLITE>_<version>

with

<TIME> is the product reference date based on the OLCI frames sensing start time. <TIME> is expressed as “YYYYMMDDhhmmss”, where YYYY, MM, DD, HH, mm and ss denote the year, the month, the day, the hour, the minutes, and the seconds, respectively.

<AREA> is the XxxYyy 10° x 10° tile (see Figure 4)

<SATELLITE> is S3A or S3B

<version> is the processing line version used to generate the TOC reflectance tiles. The version used in this document is v2.3.4.

3.2 EVALUATION FOR THE RETRIEVAL OF VEGETATION VARIABLES

Qualitative evaluation of the performance of cloud masking and other pixel identification flags for the retrieval of vegetation variables (NDVI, FAPAR, LAI, FCOVER, DMP, NPP, LSP) focuses on the IdePix cloud detection, as only OLCI bands are used for vegetation variables calculation.

Evaluation is done through visual inspection, at full resolution, of a set of Sentinel-3 TOC reflectance products. This set of images is based on a random selection, and covers a variety of cloud and snow conditions and land cover types. The images are well spread over the whole globe (in total 11 tiles: X10Y08, X17Y04, X18Y03, X19Y07, X20Y01, X20Y07, X20Y09, X22Y03, X29Y03, X30Y10 and X32Y10, see Figure 4) and over different seasons (20180701, 20181101, 20190301, 20190601), thus covering different viewing and illumination conditions and a variety of vegetation development stages.

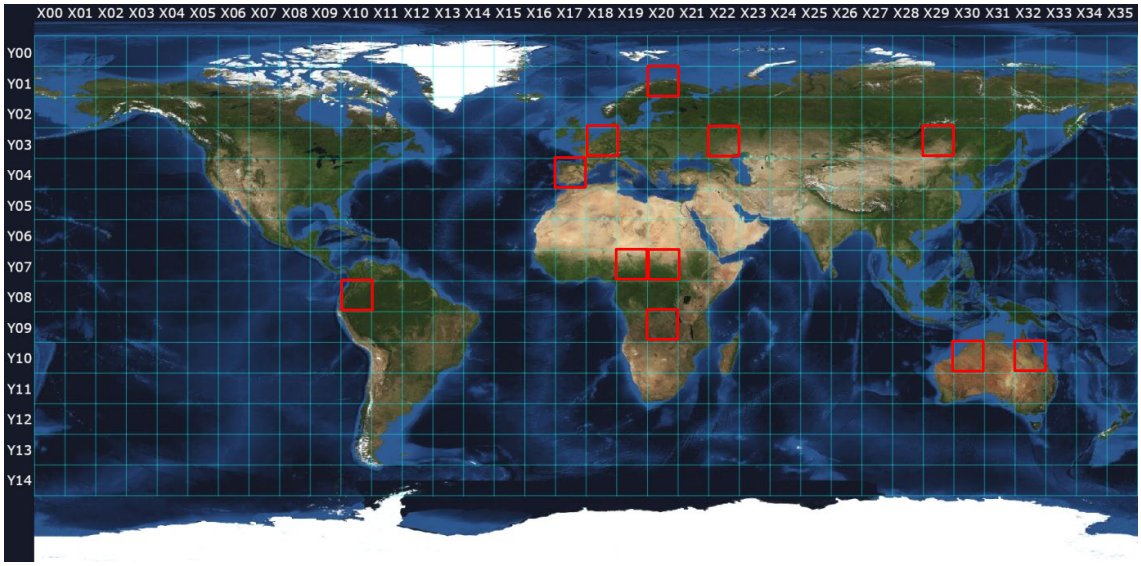

Figure 4: Grid tiles on world map[2]. Red squares indicate tiles under consideration.

For each image, a description of the performance and screenshot of the cloud, cloud shadow and snow detection is documented. This is done based on a Red-Green-Blue (RGB) (Red: Oa17_toc (865 nm), Green: Oa06_toc (560 nm), Blue: Oa03_toc (443 nm)) visualization, with overlay of the pixel identification flags.

The evaluation focuses on the retrieval of vegetation variables, i.e. on the impact of over- or under-detection of clouds, cloud shadows, snow, etc. over land surfaces, on the VNIR (i.e. OLCI) bands. The evaluation therefore focuses on the performance of IdePix on OLCI data. Although not all images can be investigated under all conditions, and as such not all cases might be covered, the analysis provides sufficient qualitative information to evaluate the pixel identification quality.

Figure 6 to Figure 64 in the next Chapter look at the performance of the IdePix classification for the retrieval of vegetation variables.

3.3 Evaluation for Burned Areas detection

For the evaluation for the BA detection, images were selected on a random basis, and cover a variety of cloud conditions and land cover types. The images were selected from tiles X10Y08, X19Y07,

[2]Background image source: http://server.arcgisonline.com/arcgis/rest/services/ESRI_Imagery_World_2D/MapServer

X20Y07, X20Y09, X22Y03, X29Y03, X30Y10 and X32Y10 (Figure 4). Dates were chosen randomly.

The BA algorithm uses OLCI and SLSTR bands and filters out cloud contaminated reflectance observations following the recommendations listed on the quality assessment report of CGLOPS TOC reflectance [CGLOPS1_QAR_TOC-reflectance300m-V2.3]. Observations are filtered out if flagged (1) as saturated by the 'quality_flags' layer of the OLCI Level 1B product, (2) as invalid, cloud, cloud shadow or snow/ice by the 'pixel_classif_flags' layer generated by IdePix (Figure 1), (3) with aerosol optical thickness >1 by 'AC_process_flag' layer generated by the atmospheric correction module (very high or critical AOT) [CGLOPS1_ATBD_S3-AC-V1.1], or (4) as cloud by 'cloud_an' layer from SLSTR Level 1B product (Figure 3).

The accuracy of the relevant quality flags for the BA algorithm, among the layers 'cloud_an' and 'pixel_classif_flags', were assessed by examining them in comparison to the RGB images of the selected tiles.

© CGLOPS Lot-1 consortium

4 RESULTS

4.1 EVALUATION FOR THE RETRIEVAL OF VEGETATION VARIABLES

The following examples show RGB composites over different land surfaces, and covering a variety of cloud and snow conditions. For each example, the Product ID is mentioned, according to the naming standard specified in Section 3.1. The zoom level differs between the figures, resulting in a varying level of detail.

The color coding used to visualize the pixel identification flags is displayed in Figure 5. The three CLOUD, CLOUD_BUFFER and CLOUD_AMBIGUOUS flags are coded with the same red color.

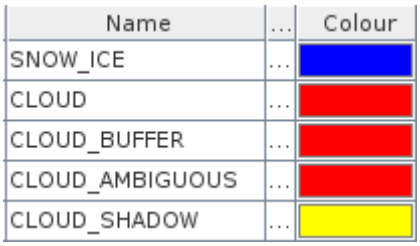

Figure 5: Color coding to visualize the pixel identification flags in the following figures

4.1.1 Detection over dark surfaces

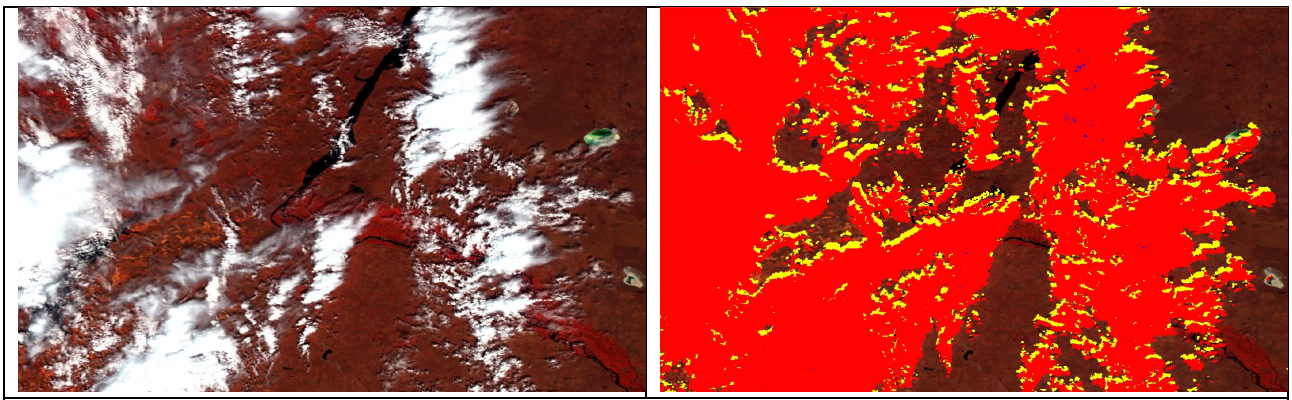

Figure 6: Product cgl_TOC_20180701072059_X22Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of cloud detection, although some over-detection of clouds and important over-detection of cloud shadows. Some bright, deep convective clouds are detected as snow/ice.

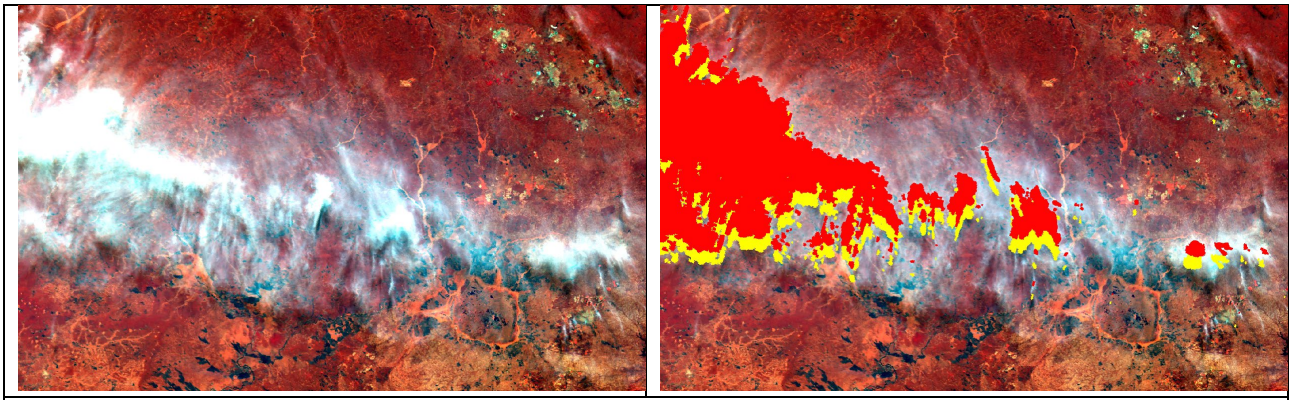

Figure 7: Product cgl_TOC_20180701073829_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows considerable under-detection of (semi-transparent) clouds and some over-detection over water and bright surfaces. Cloud shadows are underestimated.

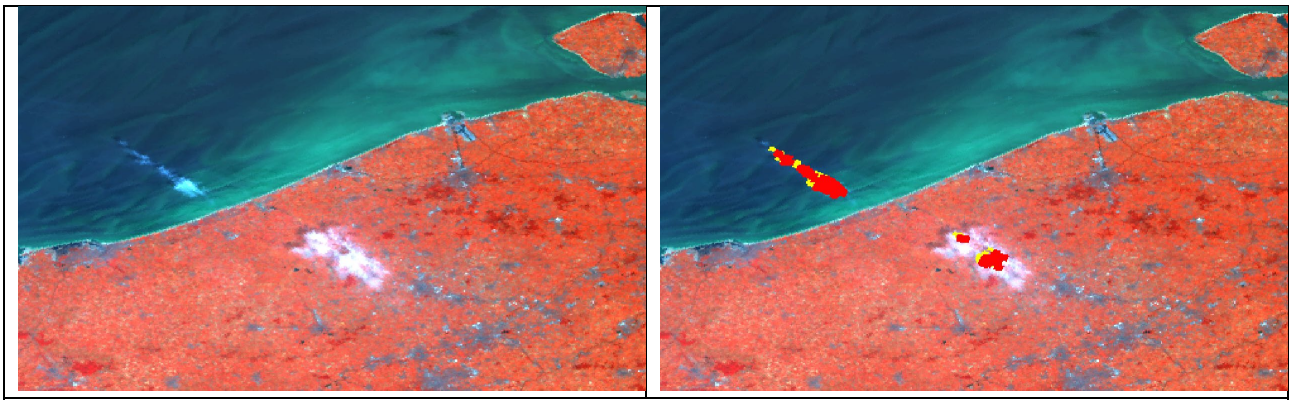

Figure 8: Product cgl_TOC_20180701103957_X18Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows considerable under-detection of (semi-transparent) clouds. Related to the under-detection of clouds, cloud shadow detection is underperforming.

© CGLOPS Lot-1 consortium

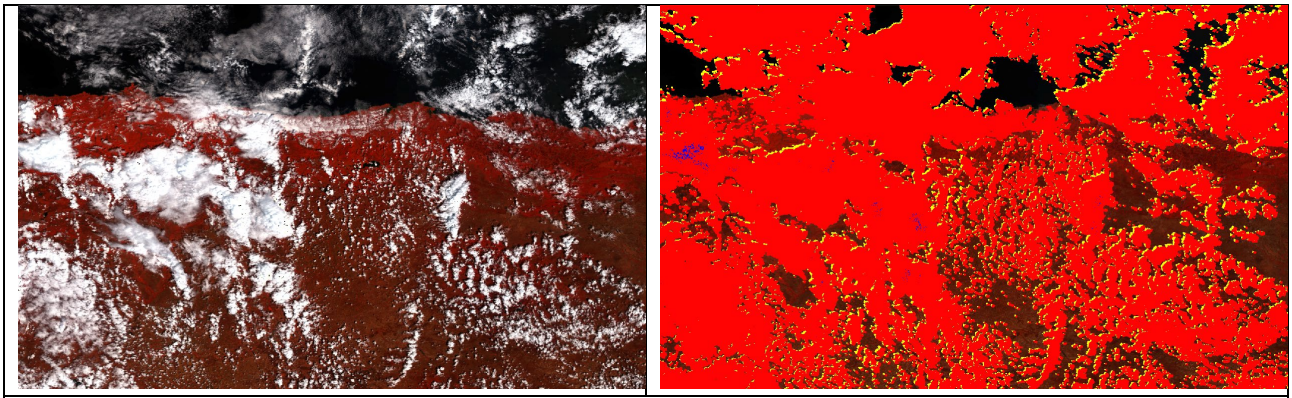

Figure 9: Product cgl_TOC_20180701104257_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the cloud and cloud shadow detection. Some bright, deep convective clouds are detected as snow/ice.

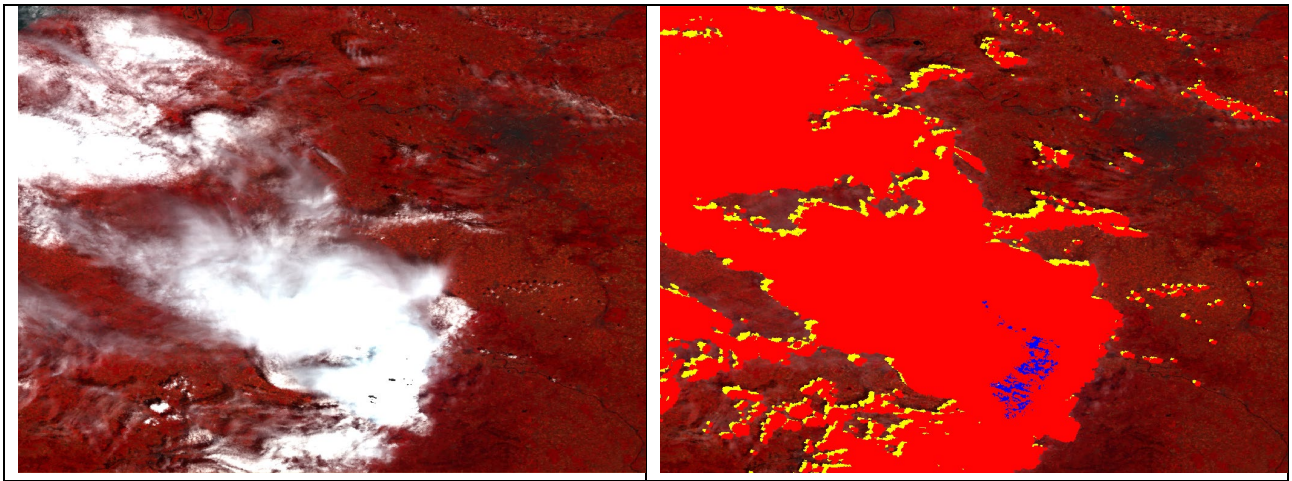

Figure 10: Product cgl_TOC_20180701104257_X18Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the cloud and cloud shadow detection. Some deep convective clouds are masked as snow/ice. Some (semi-transparent) cloud edges are not detected.

© CGLOPS Lot-1 consortium

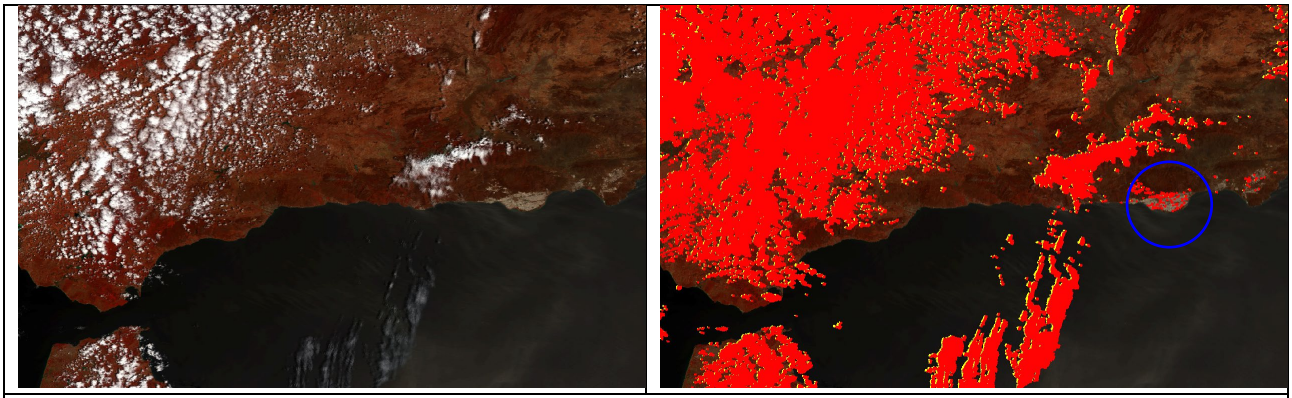

Figure 11: Product cgl_TOC_20180701104527_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the cloud and cloud shadow detection, although with some overdetection of clouds over bright clear sky surfaces (see blue circle).

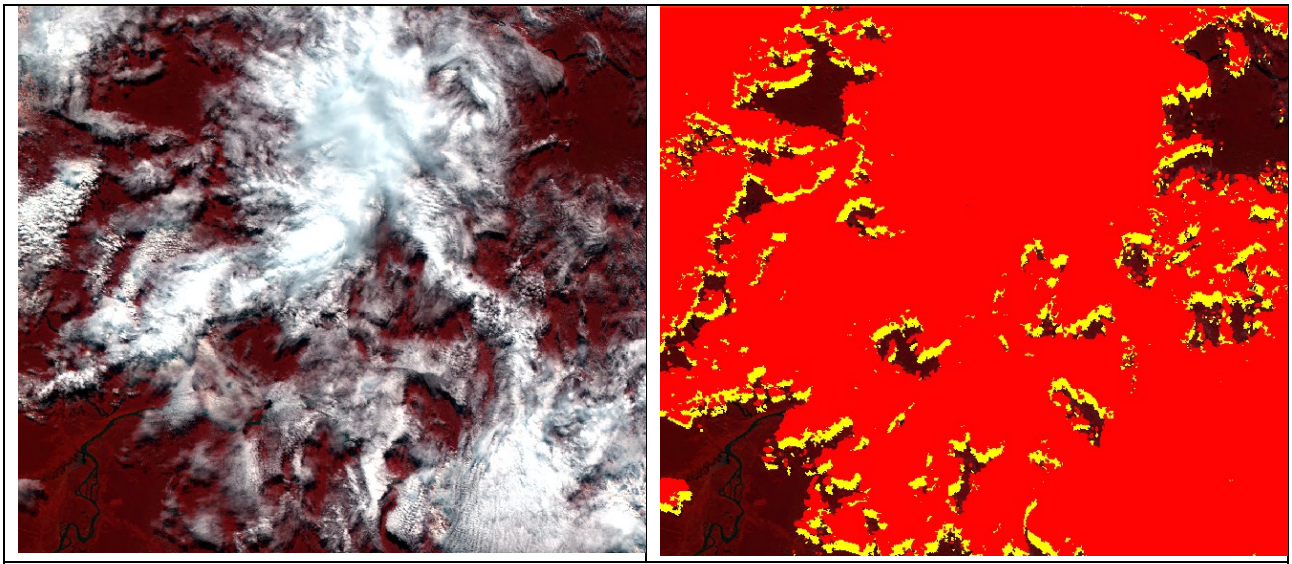

Figure 12: Product cgl_TOC_20180701141956_X10Y08_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the cloud and cloud shadow detection.

© CGLOPS Lot-1 consortium

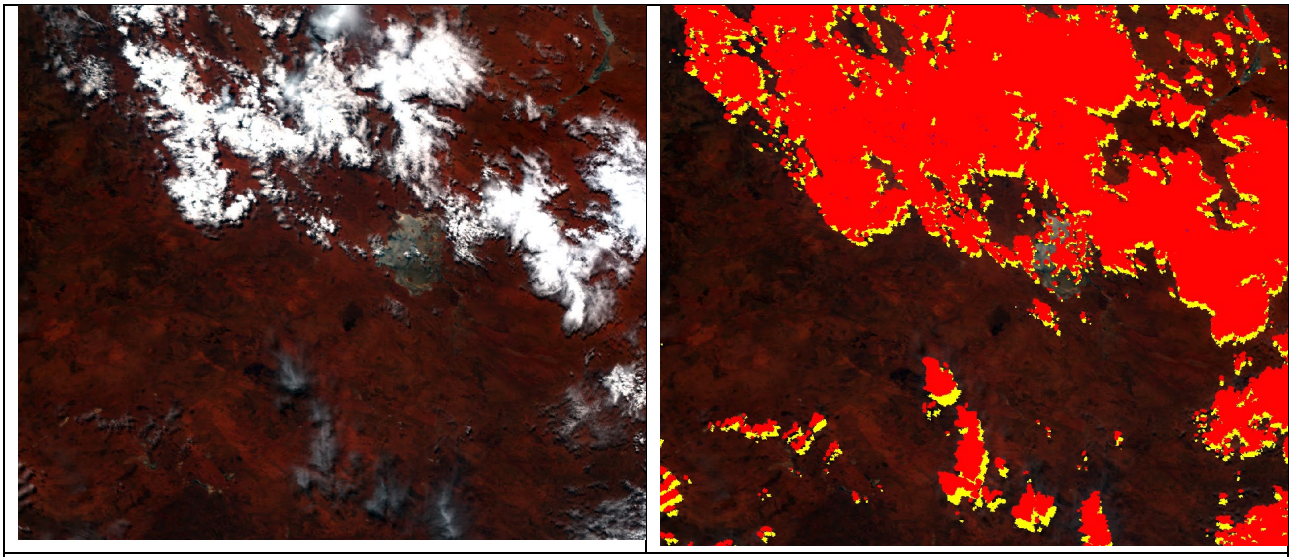

Figure 13: Product cgl_TOC_20181101010915_X30Y10_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the cloud detection, with some over-detection over bright surfaces. Cloud shadows are overestimated.

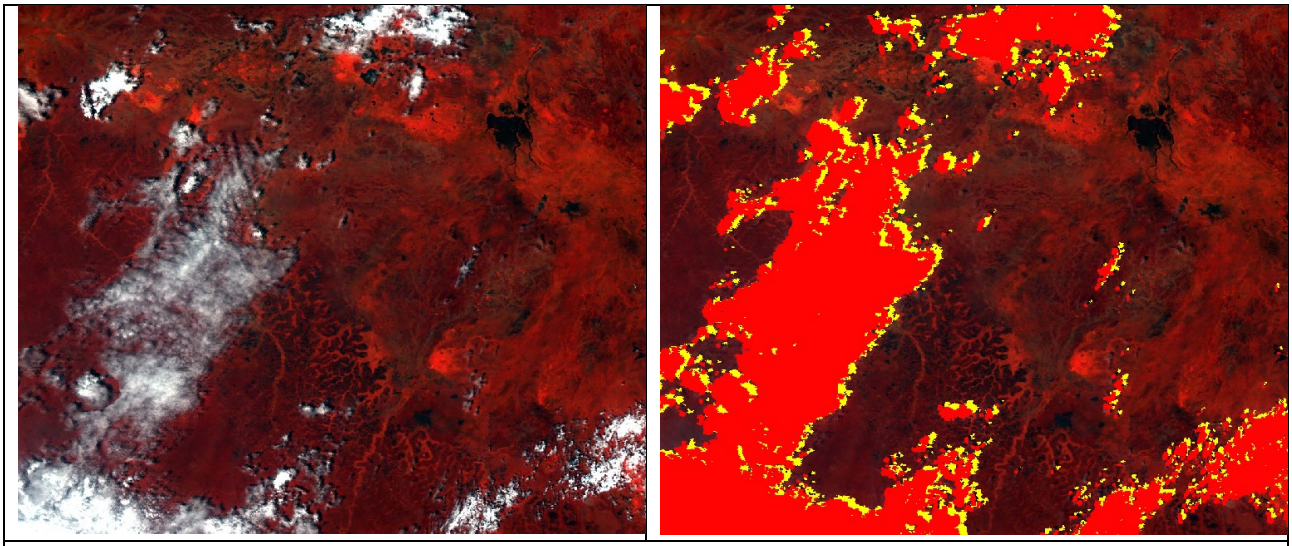

Figure 14: Product cgl_TOC_20181101074412_X20Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the cloud and cloud shadow detection.

© CGLOPS Lot-1 consortium

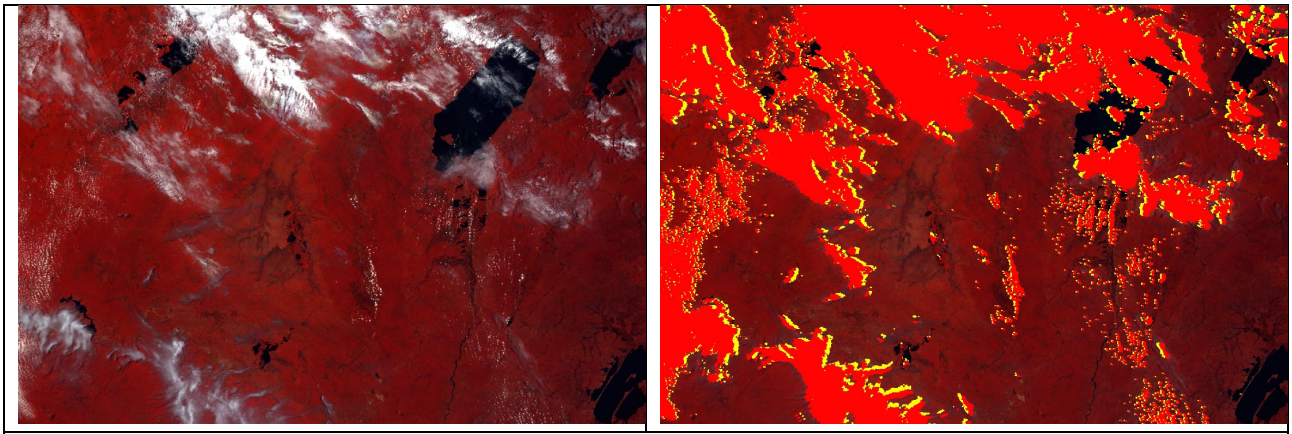

Figure 15: Product cgl_TOC_20181101074712_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of cloud detection, but large overestimation of cloud shadows.

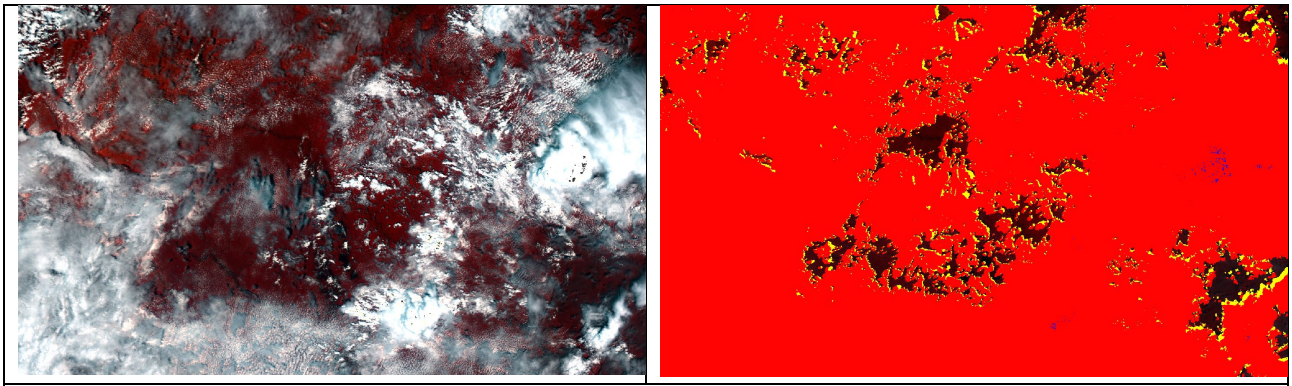

Figure 16: Product cgl_TOC_20181101142809_X10Y08_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the pixel identification, although a few dense clouds are masked as snow/ice.

© CGLOPS Lot-1 consortium

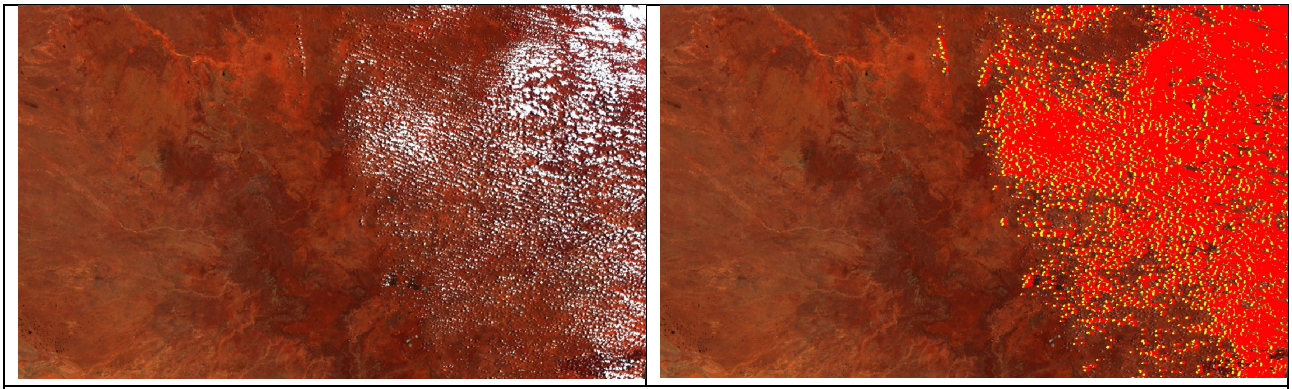

Figure 17: Product cgl_TOC_20190301001532_X32Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the pixel identification, although a number of small clouds are not identified.

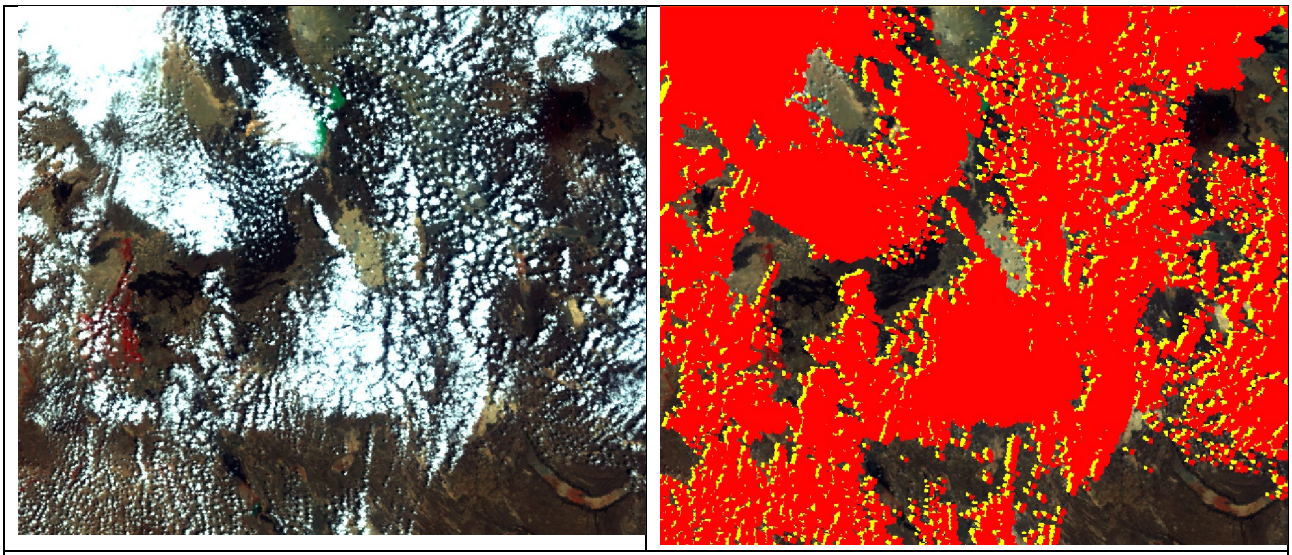

Figure 18: Product cgl_TOC_20190301072957_X22Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the pixel identification, although a number of small clouds and cloud edges are not identified.

© CGLOPS Lot-1 consortium

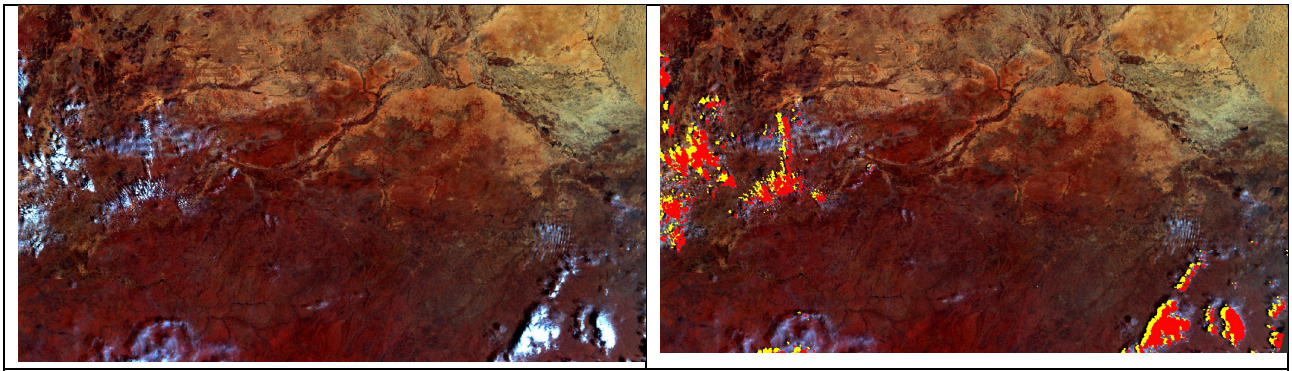

Figure 19: Product cgl_TOC_20190301083128_X20Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows some under-detection of (semi-transparent) clouds and cloud borders, and under-detection of cloud shadows.

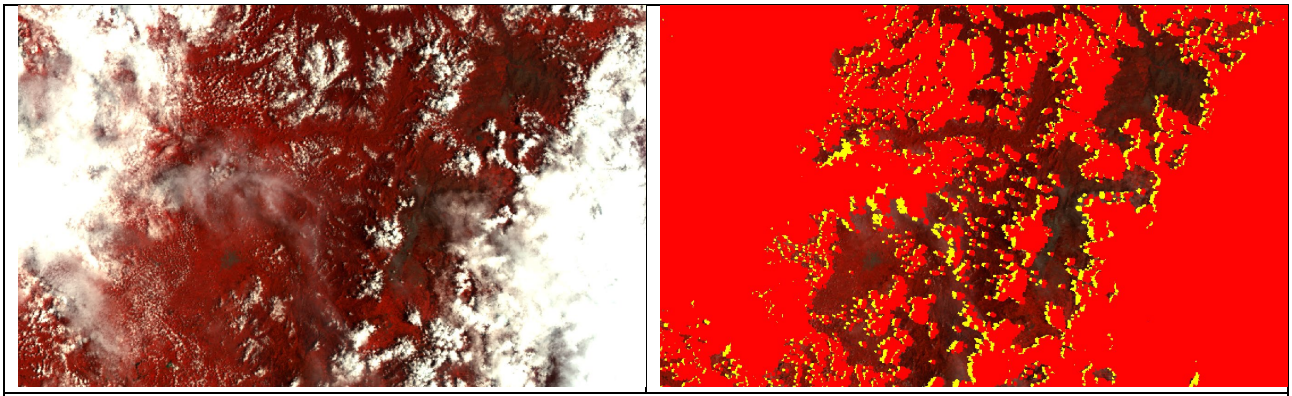

Figure 20: Product cgl_TOC_20190301151825_X10Y08_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of cloud detection, with some under-detection of (semi-transparent) cloud borders.

© CGLOPS Lot-1 consortium

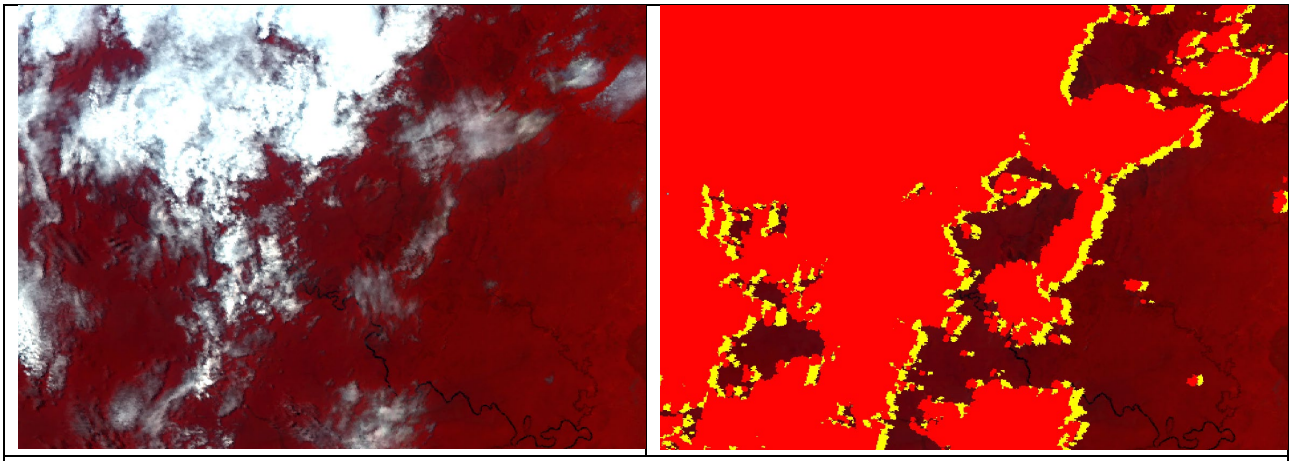

Figure 21: Product cgl_TOC_20190301141653_X10Y08_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows overestimation of clouds over water bodies and large overestimation of cloud shadows.

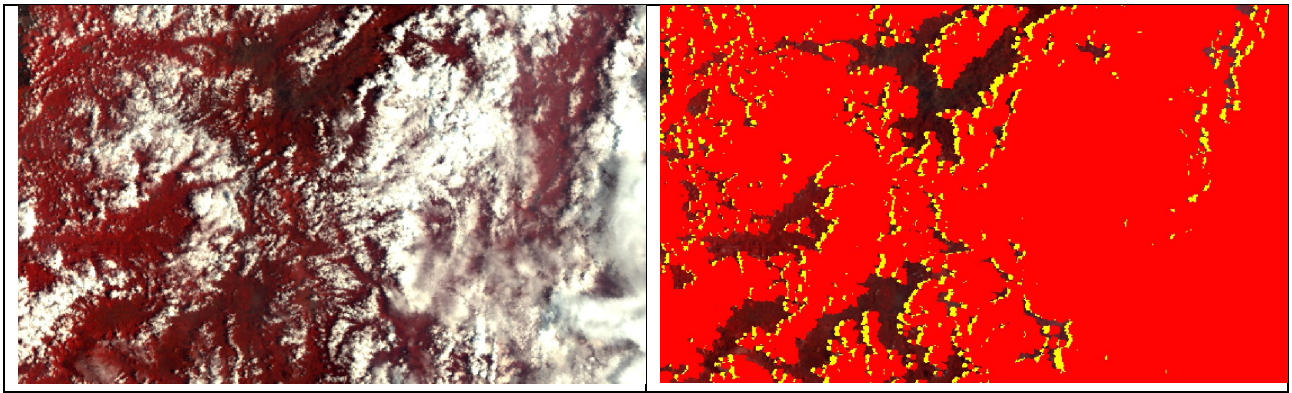

Figure 22: Product cgl_TOC_20190301152125_X10Y08_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of cloud detection.

© CGLOPS Lot-1 consortium

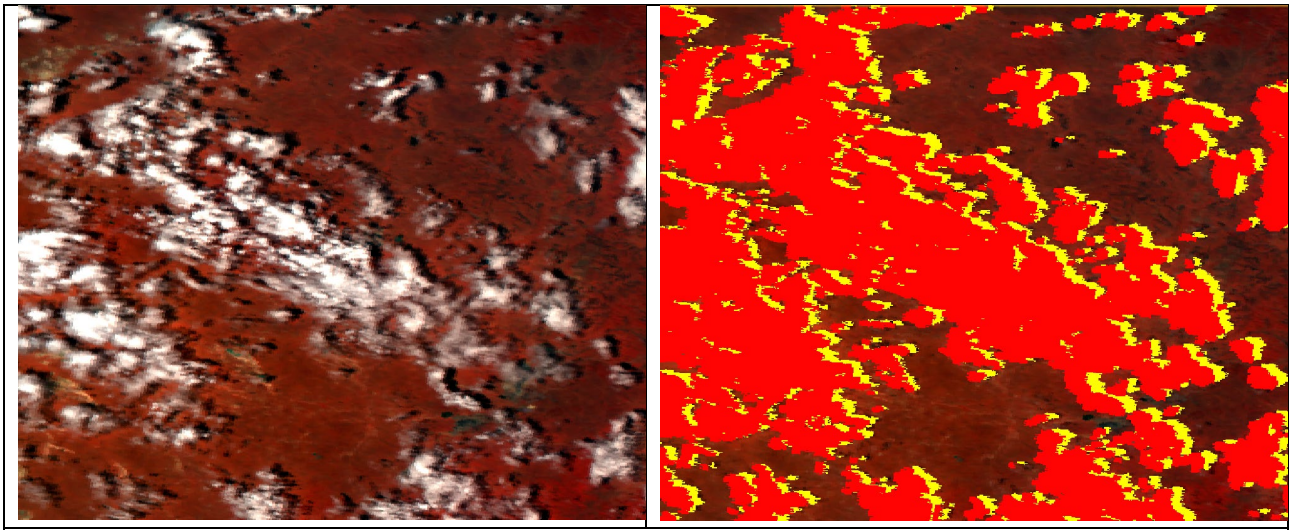

Figure 23: Product cgl_TOC_20190601015327_X29Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of cloud and cloud shadow detection.

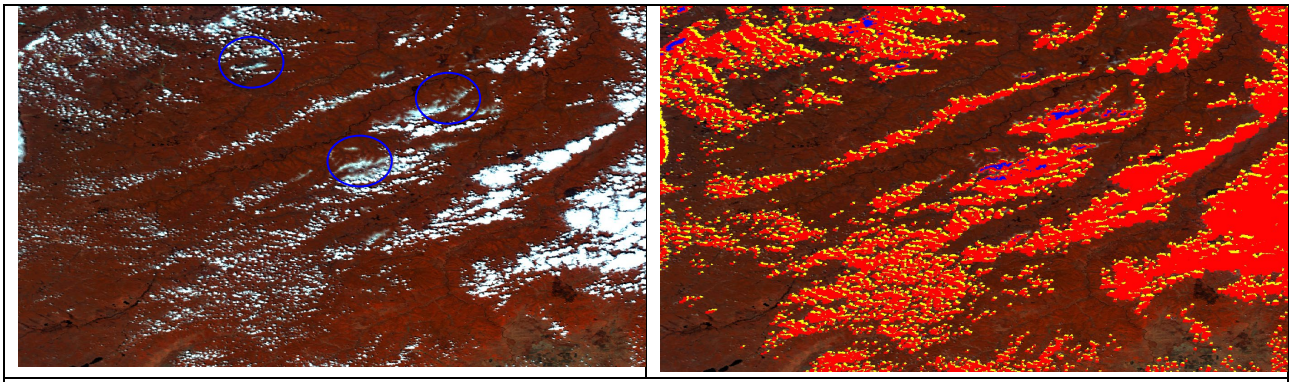

Figure 24: Product cgl_TOC_20190601023003_X29Y03_S3A_v2.3.4 (RGB Oa17, Oa06, Oa03)

This example shows overall good performance of the cloud detection, with some under-detection of small (semi-transparent) clouds. In addition, some pixels are identified as snow (see blue circles), which seems correct.

© CGLOPS Lot-1 consortium

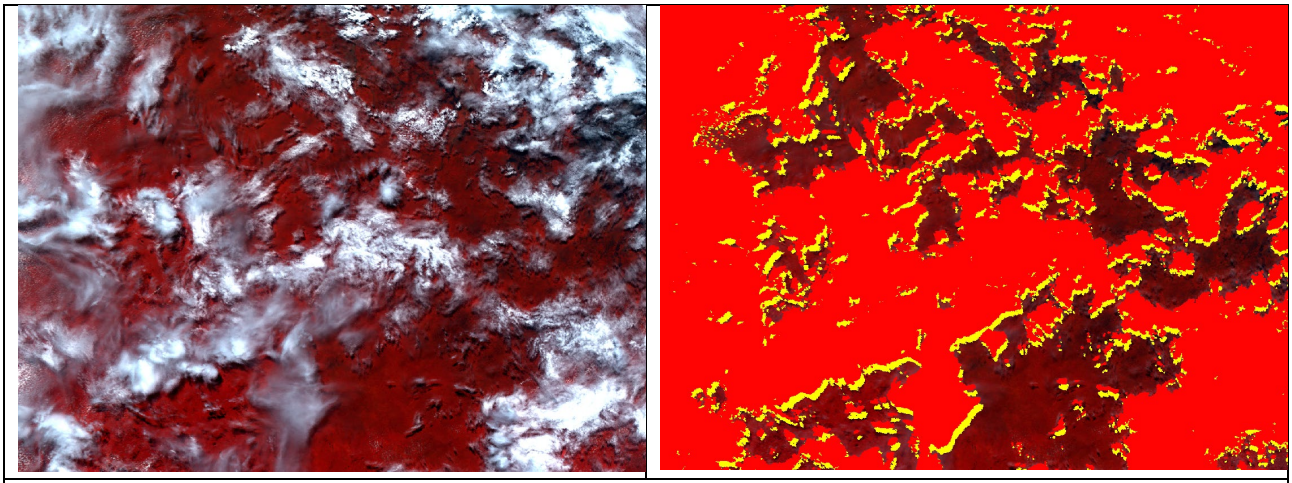

Figure 25: Product cgl_TOC_20190601074801_X20Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows relatively good performance of cloud detection and cloud shadow detection.

4.1.2 Detection over bright surfaces

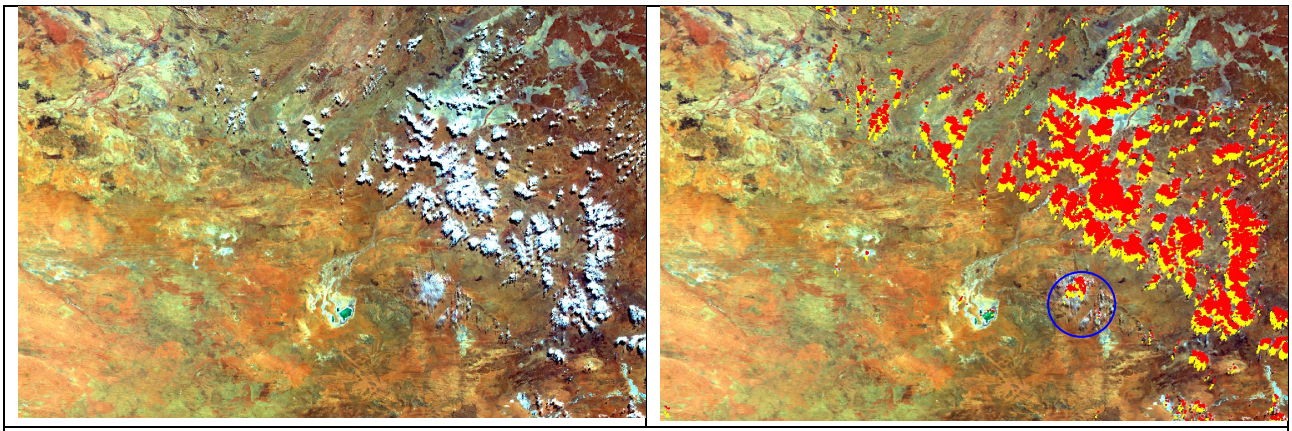

Figure 26: Product cgl_TOC_20180701005432_X30Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of cloud detection, although slight over-detection over bright surface, and slight under-detection of clouds (indicated by blue circle).

© CGLOPS Lot-1 consortium

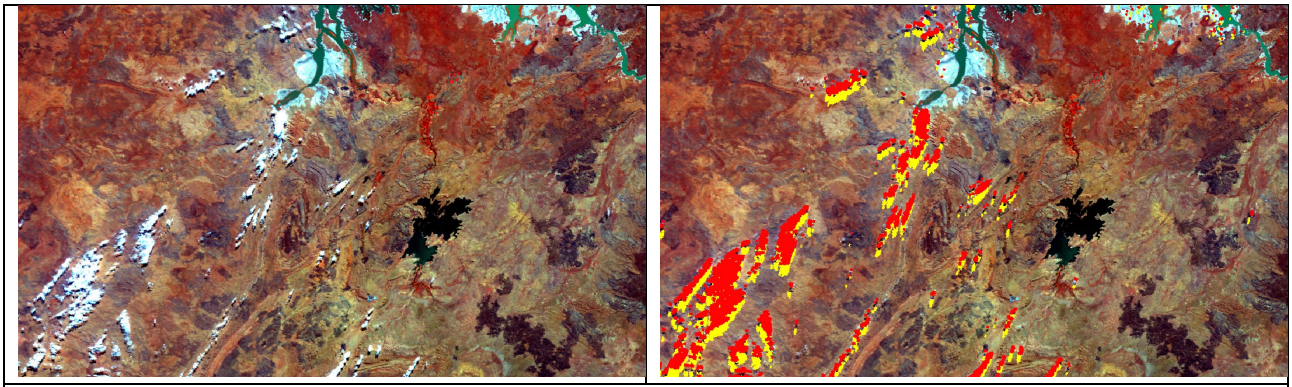

Figure 27: Product cgl_TOC_20180701005432_X30Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of cloud detection, although with slight over-detection over bright surfaces, slight under-detection of (semi-transparent) cloud borders, and overestimation of cloud shadows.

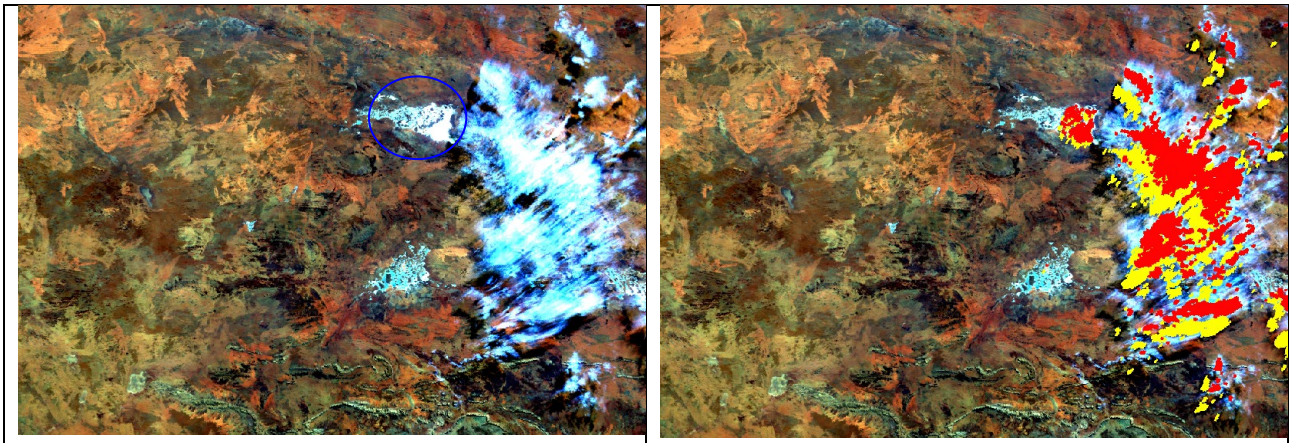

Figure 28: Product cgl_TOC_20180701005732_X30Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows considerable over-detection of clouds over bright surfaces (identified by blue circle on the left image), and under-detection of (semi-transparent) cloud borders.

© CGLOPS Lot-1 consortium

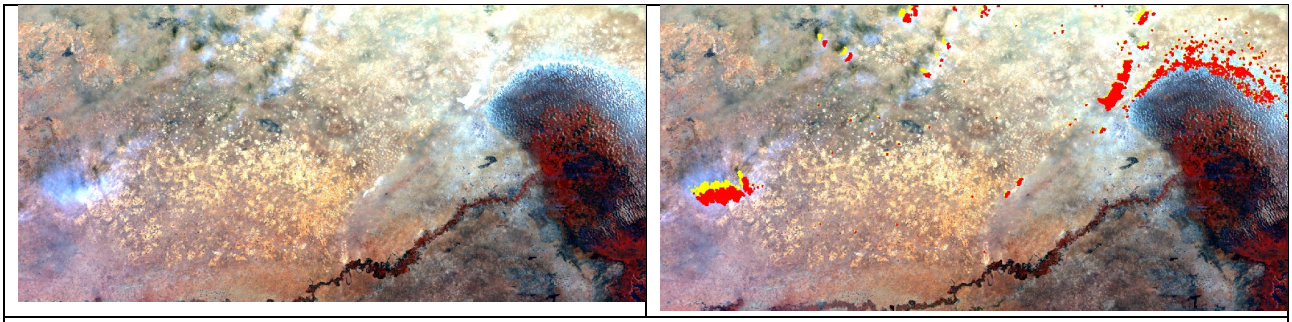

Figure 29: Product cgl_TOC_20181101092211_X19Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows considerable under-detection of semi-transparent clouds, and over-detection of clouds over bright (clear-sky) surfaces. Cloud shadows are underestimated.

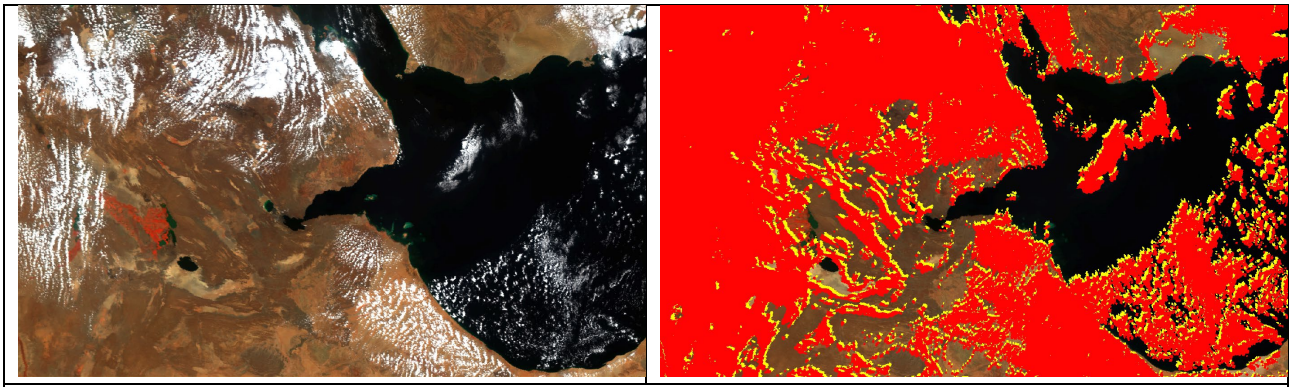

Figure 30: Product cgl_TOC_20190301065029_X22Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows wide scale over-detection of clouds and cloud shadows (none are visible on the left image) over bright surfaces.

© CGLOPS Lot-1 consortium

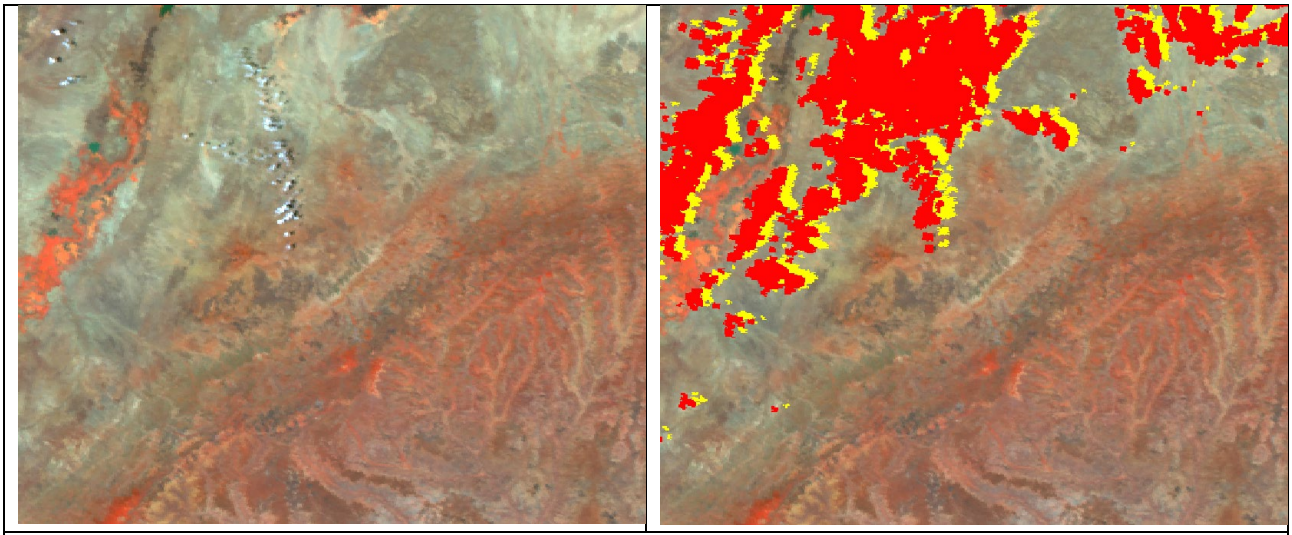

Figure 31: Product cgl_TOC_20190301065329_X22Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows considerable over-detection of clouds and cloud shadows over bright surfaces.

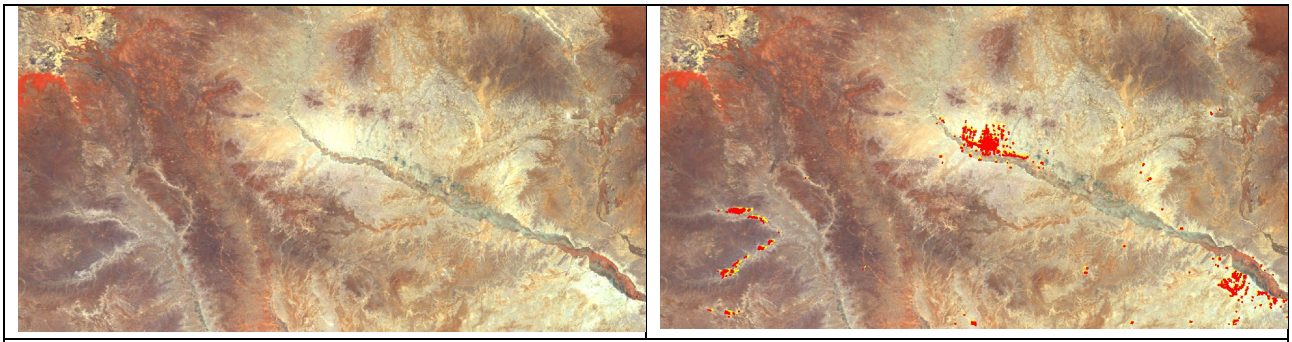

Figure 32: Product cgl_TOC_20190301065329_X22Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows considerable over-detection of clouds and cloud shadows over bright surfaces.

© CGLOPS Lot-1 consortium

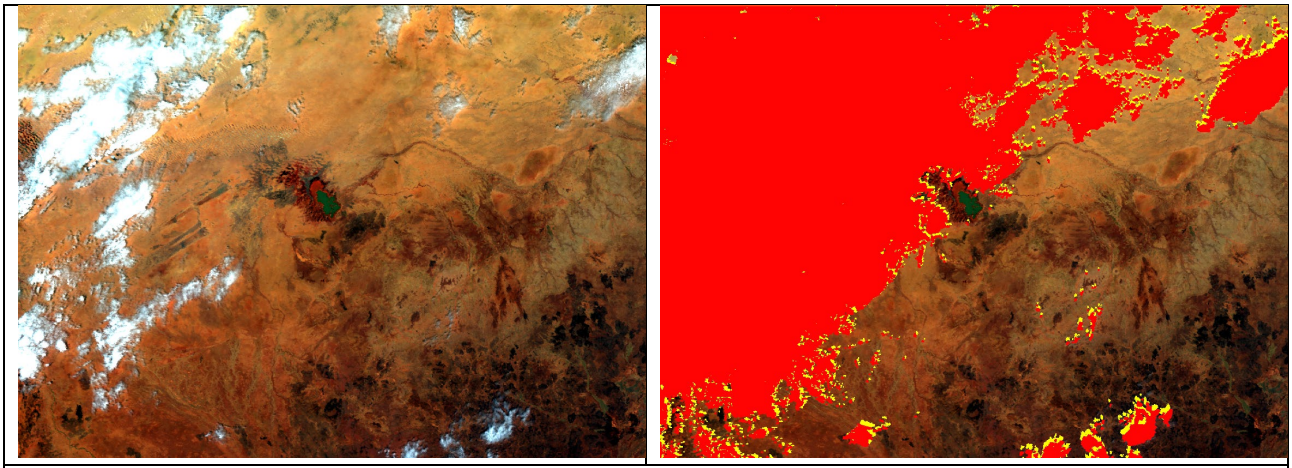

Figure 33: Product cgl_TOC_20190301083128_X19Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows large scale over-detection of clouds over bright surfaces. Cloud shadows are overestimated.

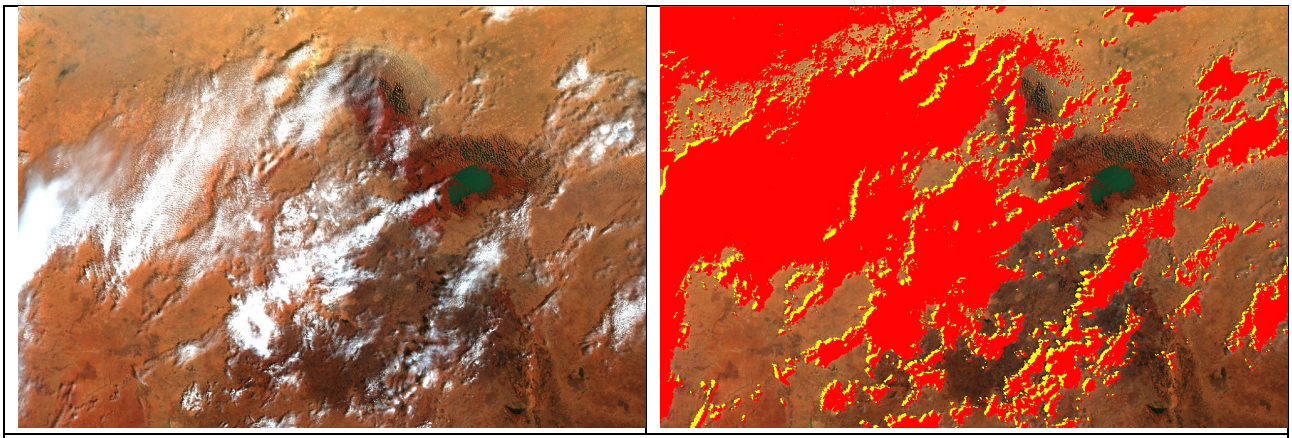

Figure 34: Product cgl_TOC_20190301091056_X19Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good detection of clouds and cloud shadows.

© CGLOPS Lot-1 consortium

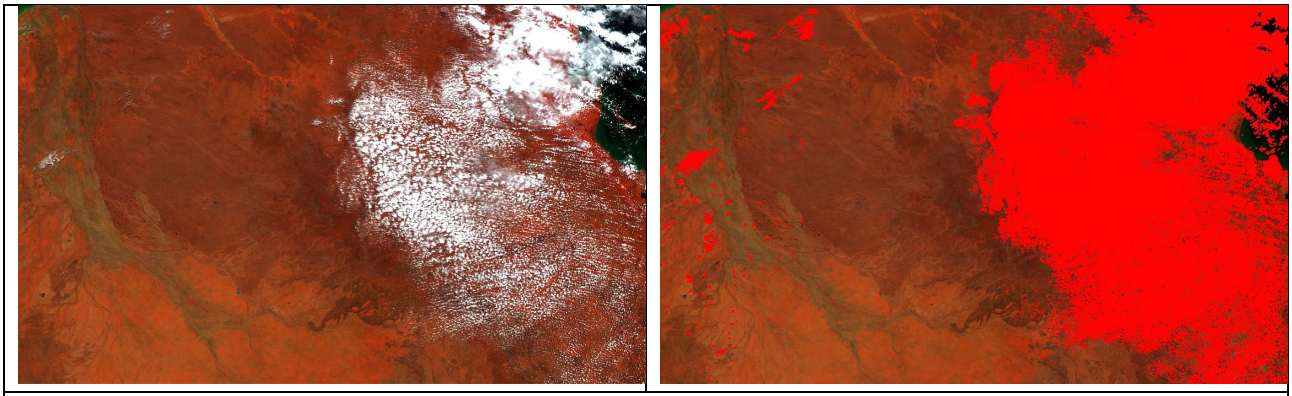

Figure 35: Product cgl_TOC_20190301234921_X32Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good detection of clouds, but with over-detection over bright surfaces.

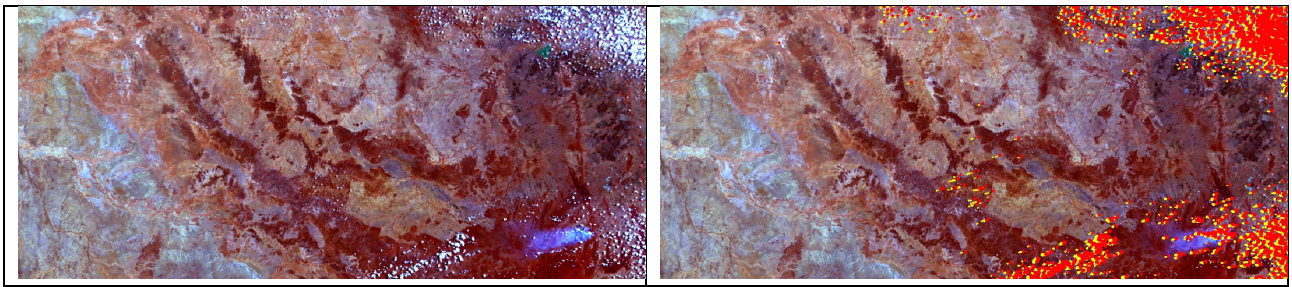

Figure 36: Product cgl_TOC_20190301235221_X32Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows some under-detection of (small) clouds and a smoke plume.

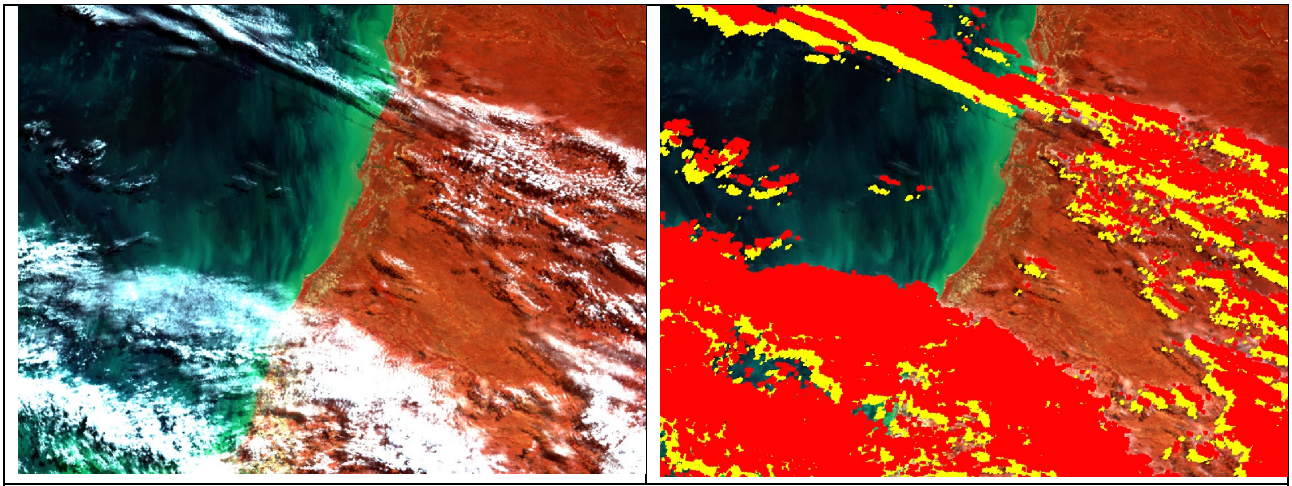

Figure 37: Product cgl_TOC_20190601003027_X32Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows under-detection of small (semi-transparent) clouds and cloud edges.

© CGLOPS Lot-1 consortium

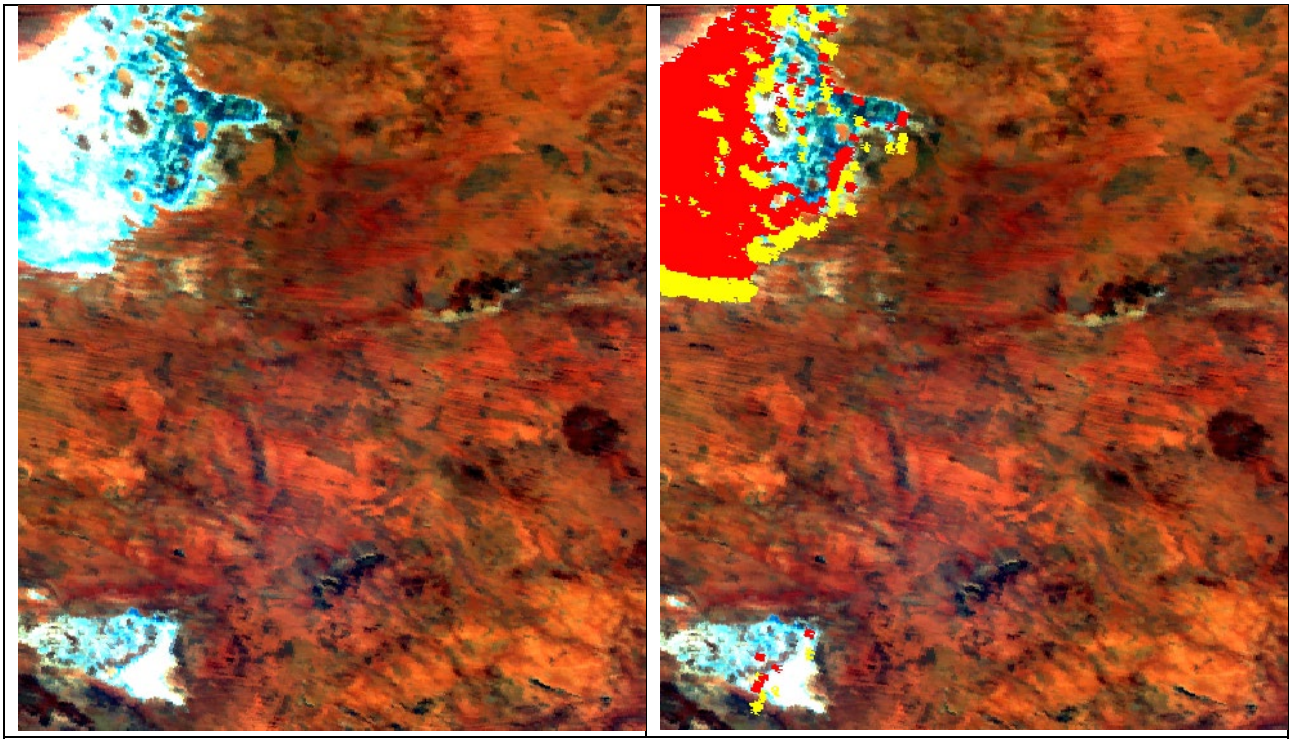

Figure 38: Product cgl_TOC_20190601003327_X30Y10_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows over-detection of clouds over very bright surfaces.

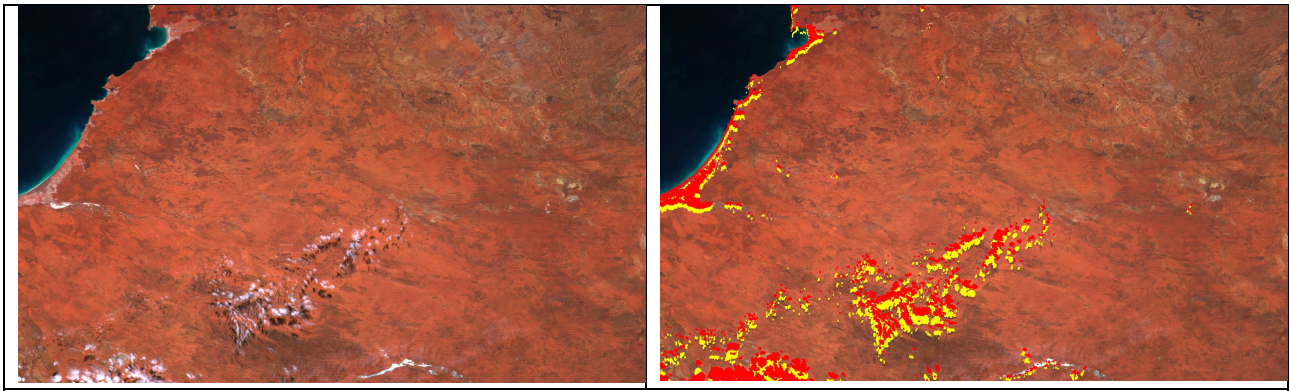

Figure 39: Product cgl_TOC_20190601011004_X30Y10_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows over-detection of clouds over very bright surfaces, and under-detection of small clouds and (semi-transparent) cloud edges.

© CGLOPS Lot-1 consortium

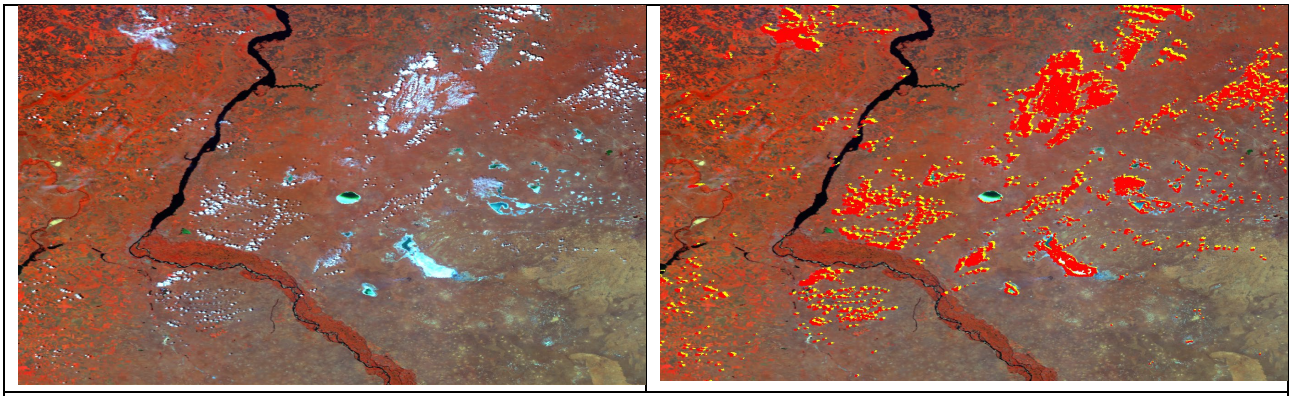

Figure 40: Product cgl_TOC_20190601065624_X22Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows over-detection of clouds over clear sky bright surfaces.

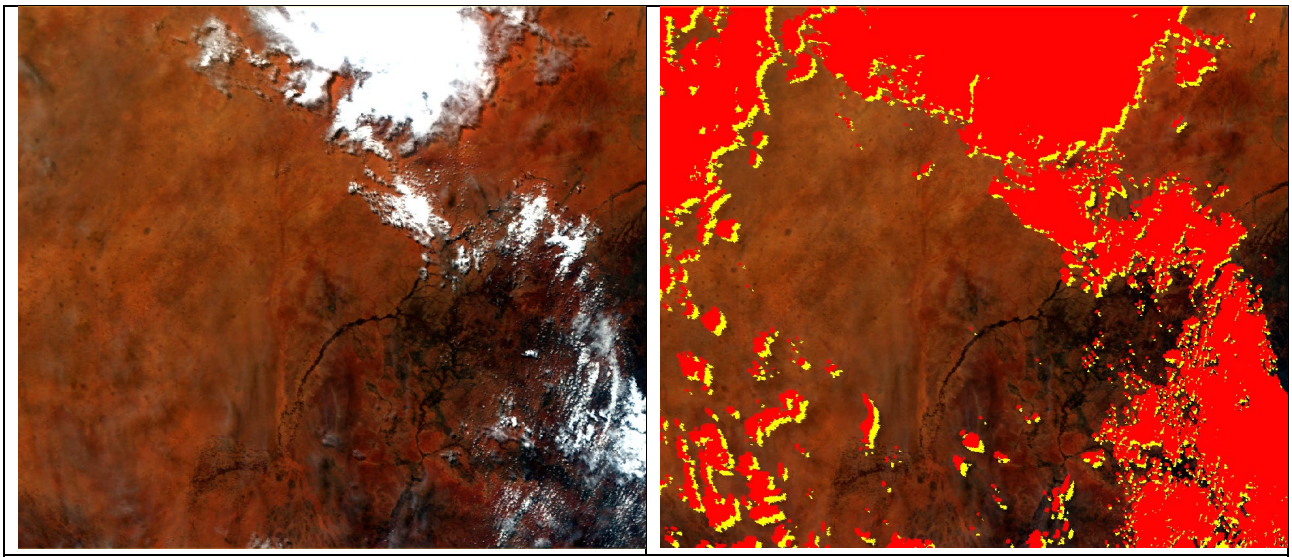

Figure 41: Product cgl_TOC_20190601074501_X20Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good detection of clouds.

© CGLOPS Lot-1 consortium

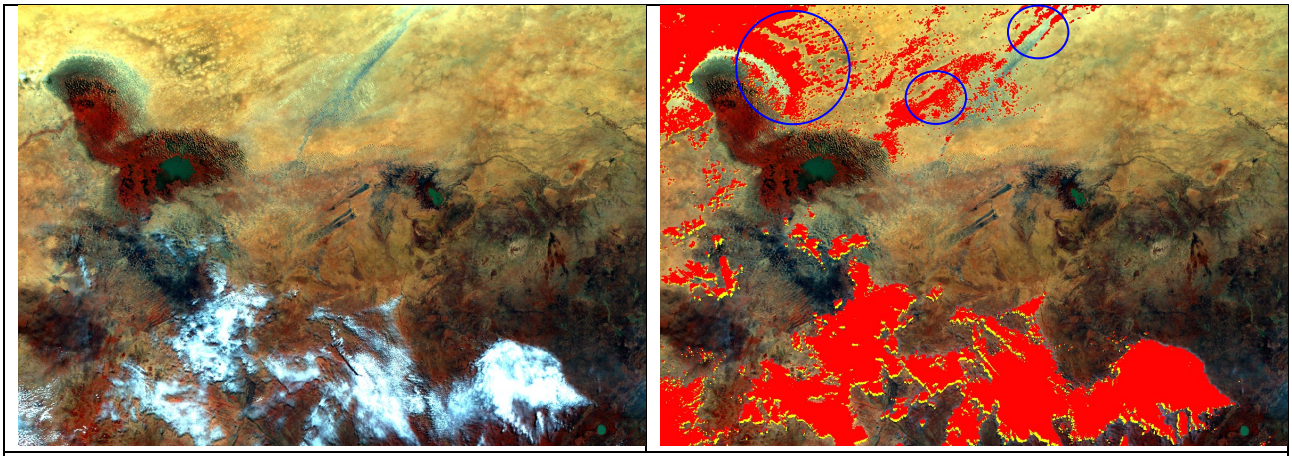

Figure 42: Product cgl_TOC_20190601084624_X19Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example overall shows good detection of clouds, although clear sky pixels are wrongly detected as cloud (see blue circles).

4.1.3 Semi-transparent clouds

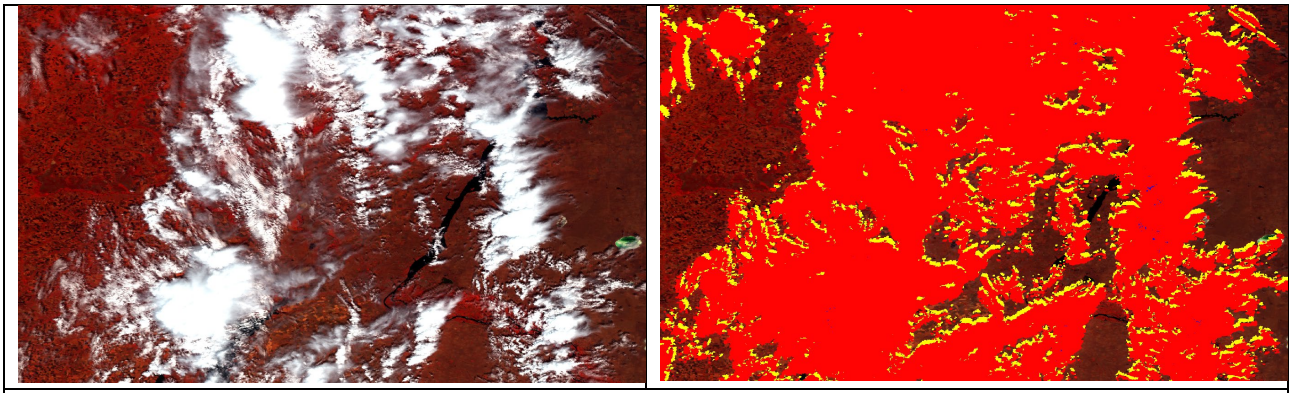

Figure 43: Product cgl_TOC_20180701072029_X22Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of cloud detection, also for semi-transparent clouds, although important over-detection of cloud shadows.

© CGLOPS Lot-1 consortium

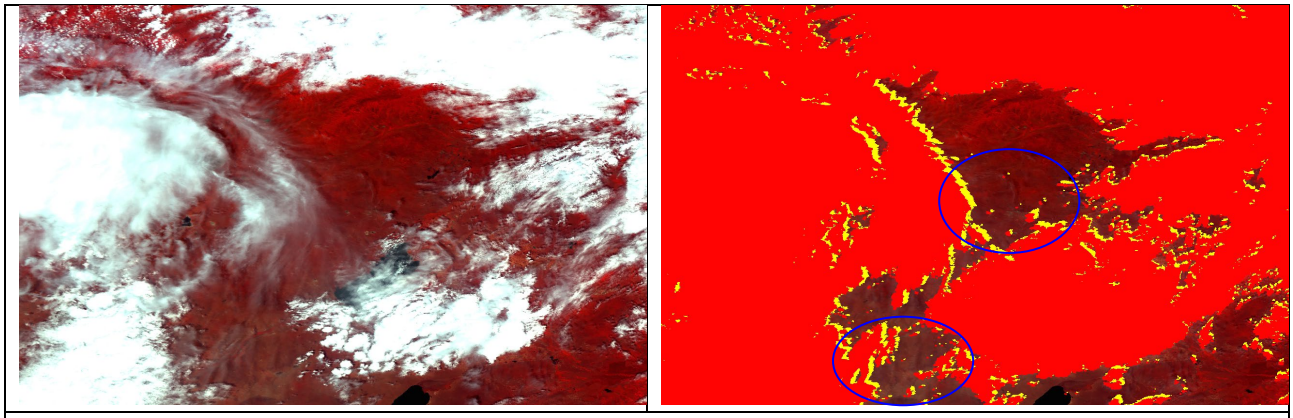

Figure 44: Product cgl_TOC_20180701021731_X29Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of cloud detection, but some under-detection of semi-transparent clouds (blue circles).

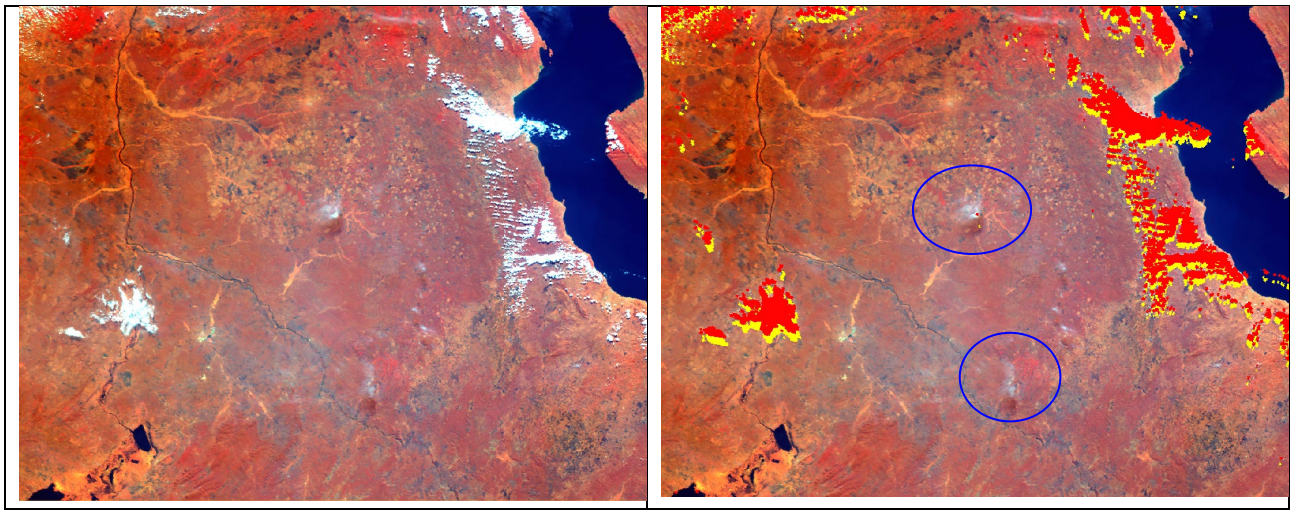

Figure 45: Product cgl_TOC_20180701073559_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows important under-detection of semi-transparent clouds.

© CGLOPS Lot-1 consortium

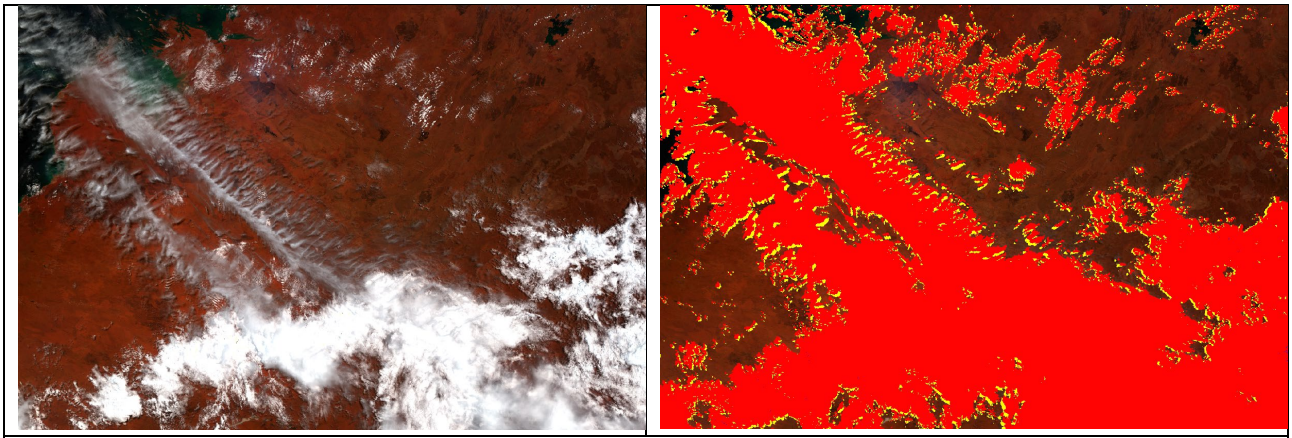

Figure 46: Product cgl_TOC_20181101010615_X30Y10_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the cloud detection, also for the semi-transparent clouds.

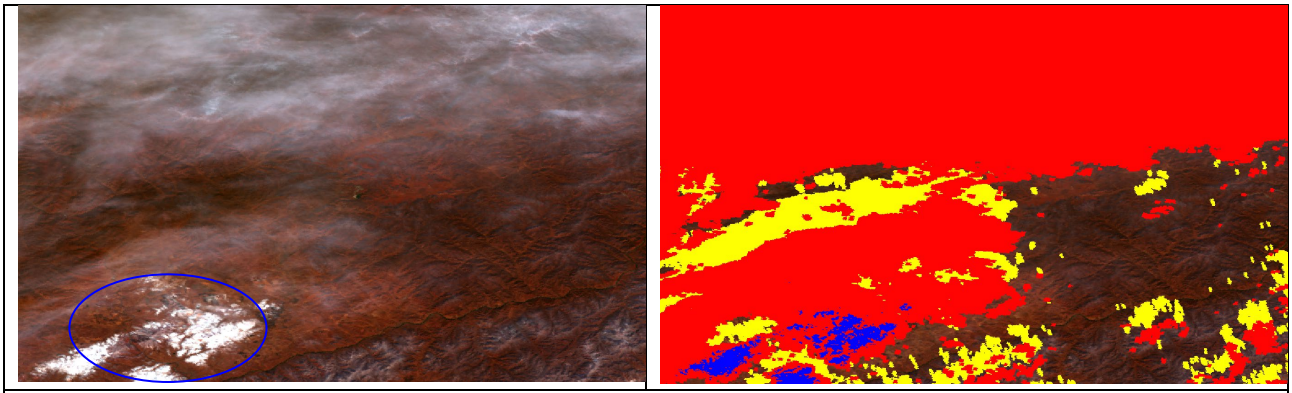

Figure 47: Product cgl_TOC_20181101022614_X29Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the cloud detection over semi-transparent clouds. There is over-detection of clouds over snow. Also cloud shadows are over-estimated.

© CGLOPS Lot-1 consortium

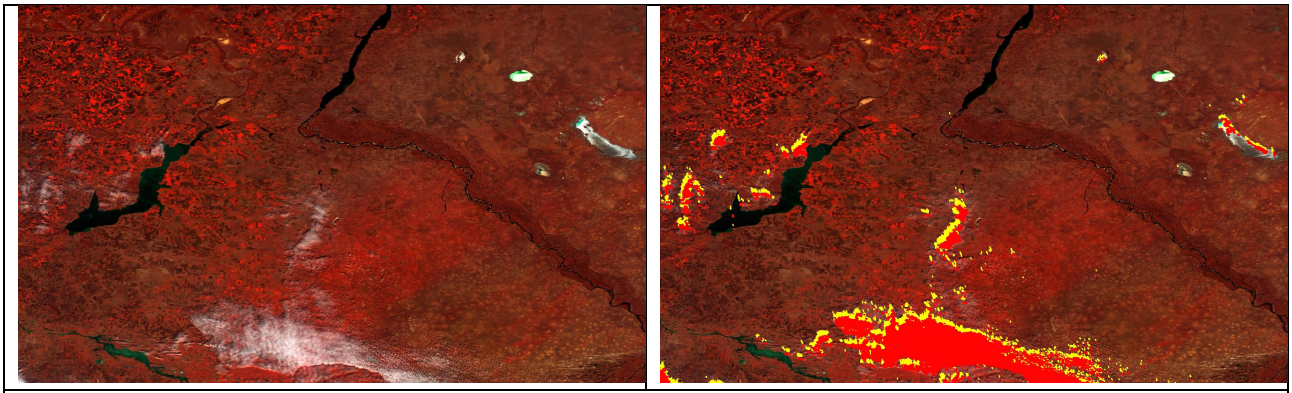

Figure 48: Product cgl_TOC_20181101073212_X22Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows some under-detection of semi-transparent clouds. Cloud shadows are over-detected (blue circles). Also, clear sky bright pixels are wrongly identified as cloud.

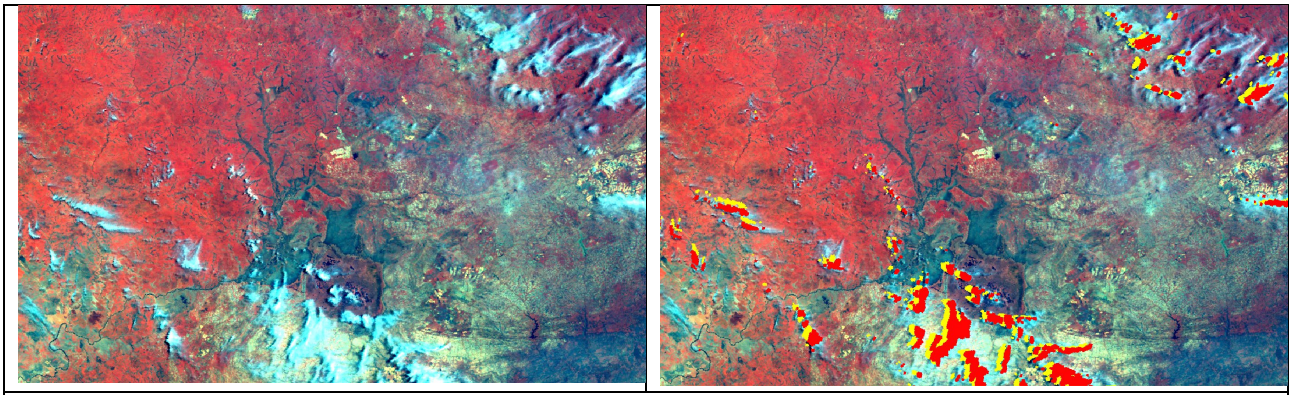

Figure 49: Product cgl_TOC_20181101075012_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows large under-detection of semi-transparent clouds.

© CGLOPS Lot-1 consortium

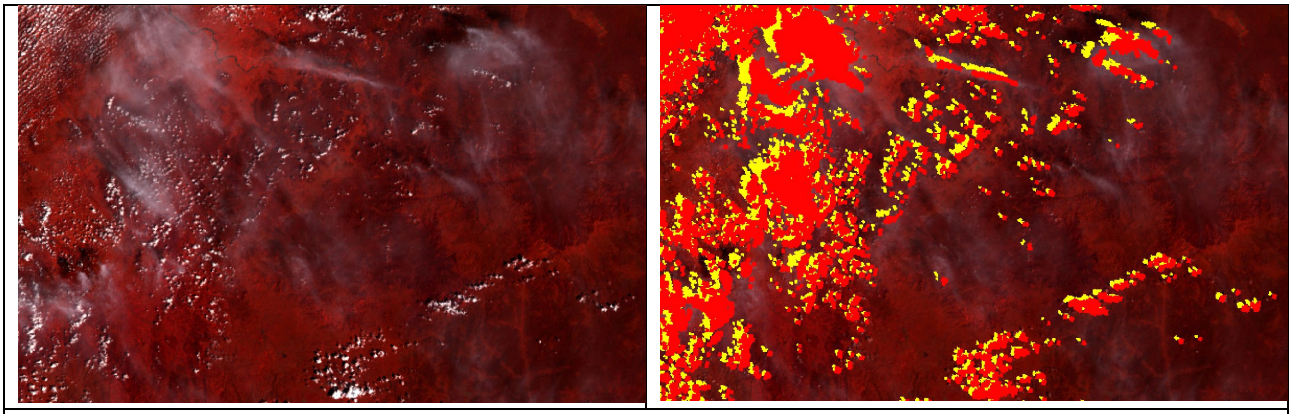

Figure 50: Product cgl_TOC_20181101092511_X19Y07_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows considerable under-detection of semi-transparent clouds over dark surfaces.

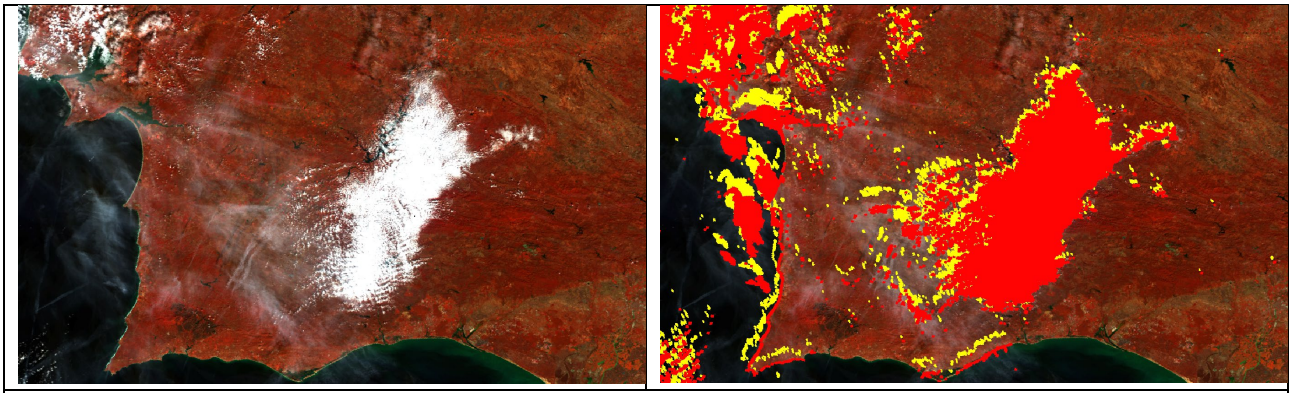

Figure 51: Product cgl_TOC_20181101105711_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows considerable under-detection of semi-transparent clouds, over-detection of cloud shadows, and over-detection of clouds at the shoreline.

© CGLOPS Lot-1 consortium

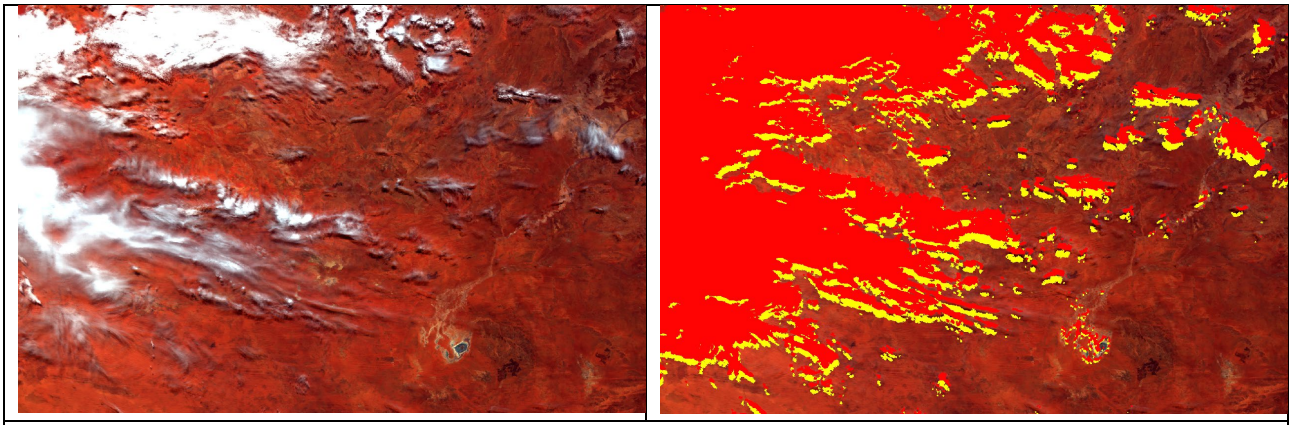

Figure 52: Product cgl_TOC_20190301005500_X30Y10_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows some under-detection of semi-transparent clouds and cloud edges, and over-detection of clouds over bright surfaces.

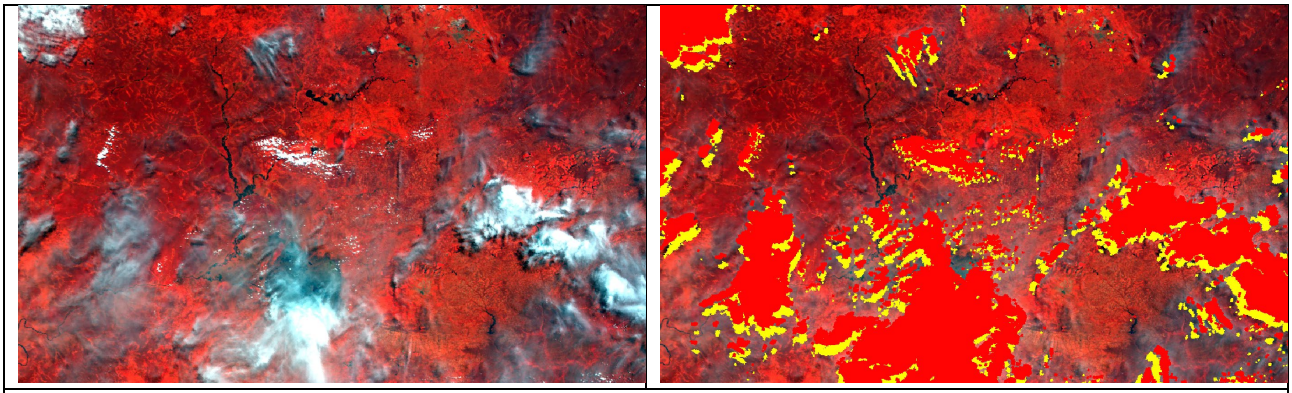

Figure 53: Product cgl_TOC_20190301073857_X20Y09_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows under-detection of semi-transparent clouds, and over-detection of cloud shadows.

© CGLOPS Lot-1 consortium

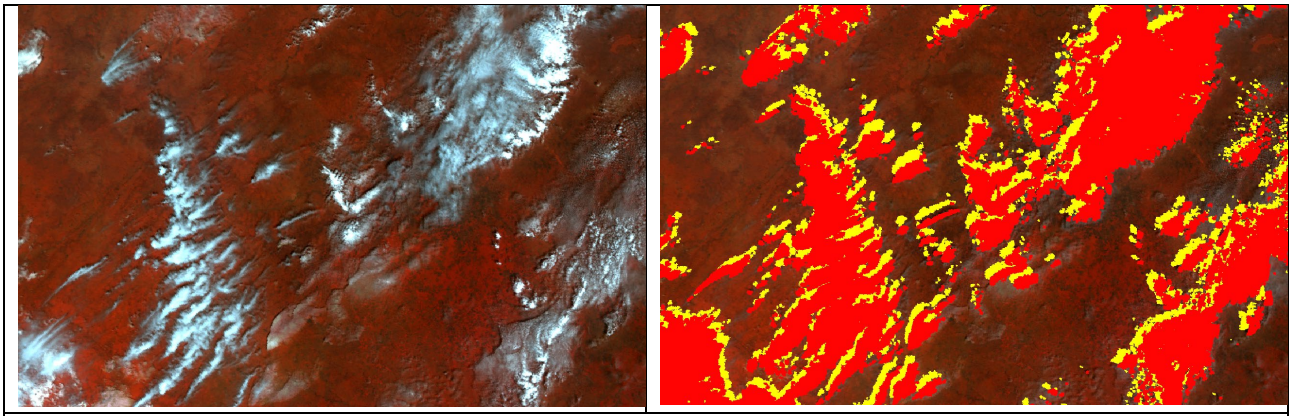

Figure 54: Product cgl_TOC_20190301083428_X19Y07_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows under-detection of semi-transparent clouds, and over-detection of cloud shadows.

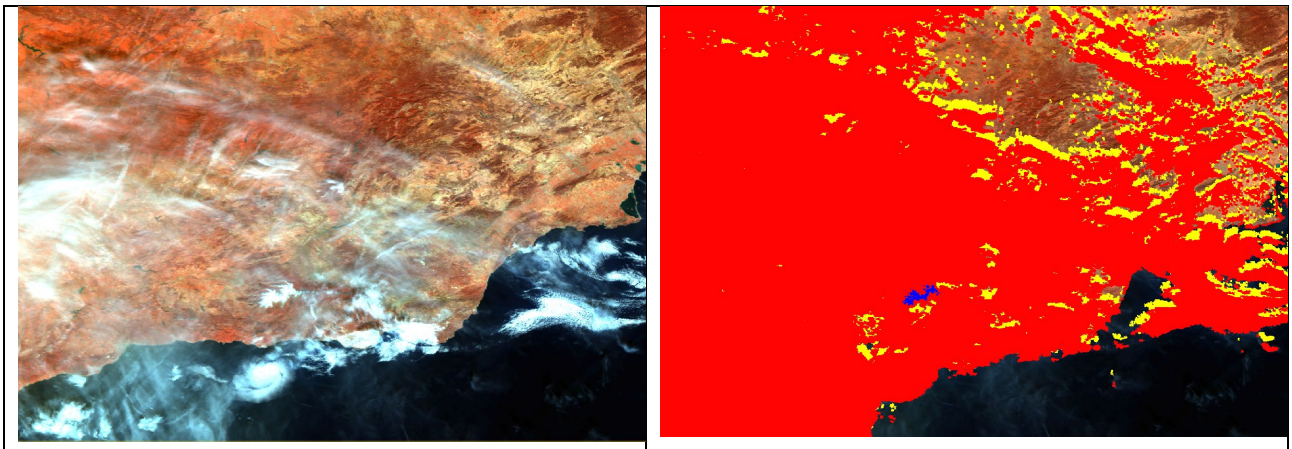

Figure 55: Product cgl_TOC_20190301100627_X17Y04_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows correct detection of semi-transparent clouds.

© CGLOPS Lot-1 consortium

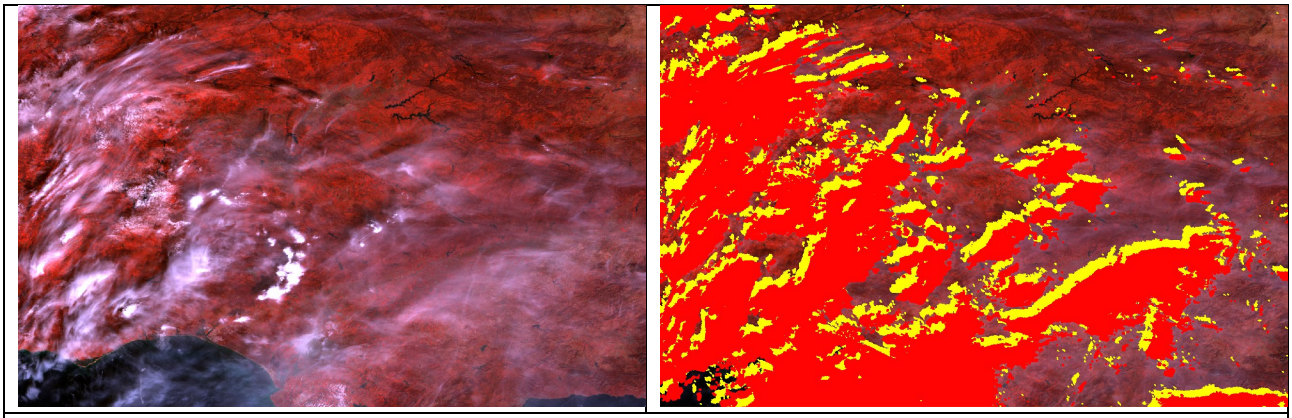

Figure 56: Product cgl_TOC_20190301104555_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows under-detection of semi-transparent clouds.

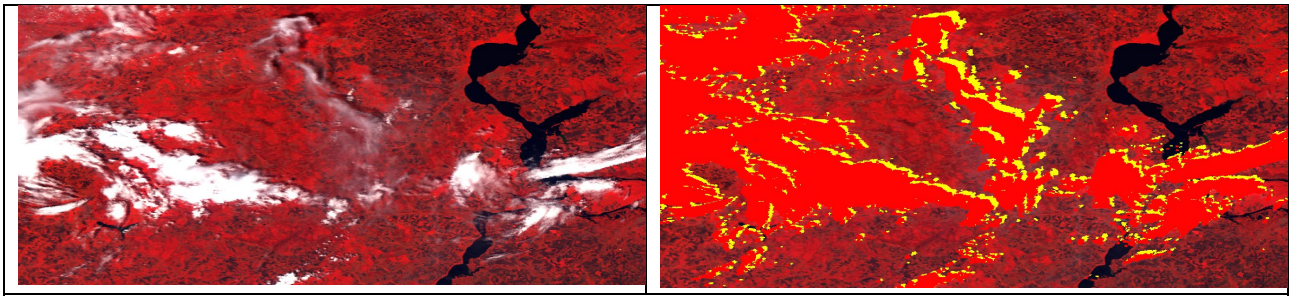

Figure 57: Product cgl_TOC_20190601065324_X22Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of the cloud and cloud shadow detection.

© CGLOPS Lot-1 consortium

4.1.4 Snow detection

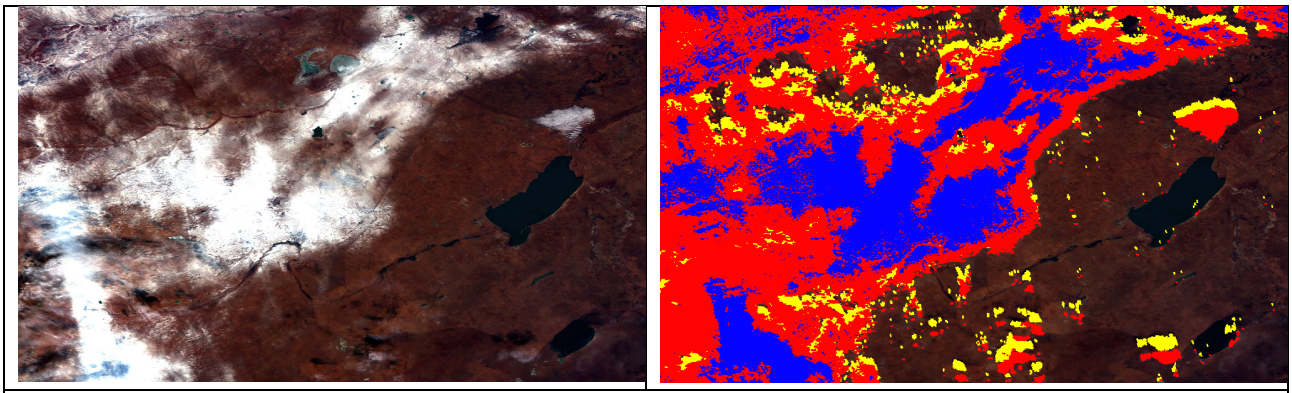

Figure 58: Product cgl_TOC_20181101022914_X29Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of snow/ice detection in combination with semi-transparent clouds.

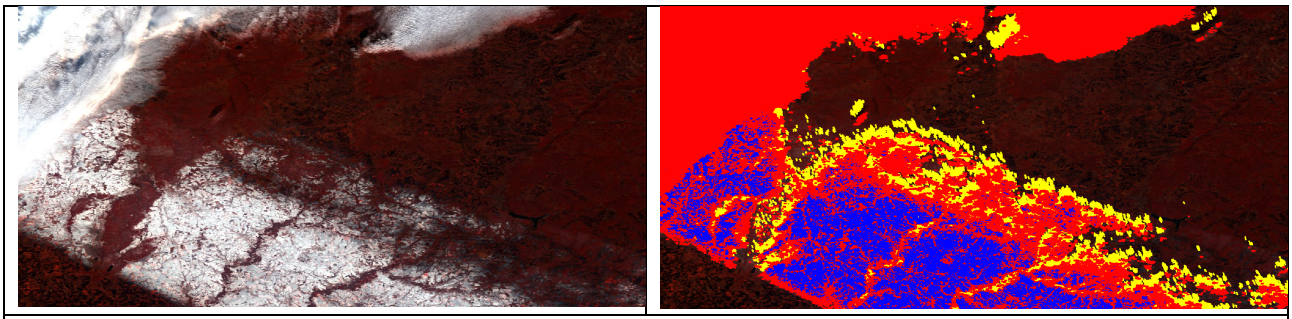

Figure 59: Product cgl_TOC_20181101072912_X22Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of snow/ice detection.

© CGLOPS Lot-1 consortium

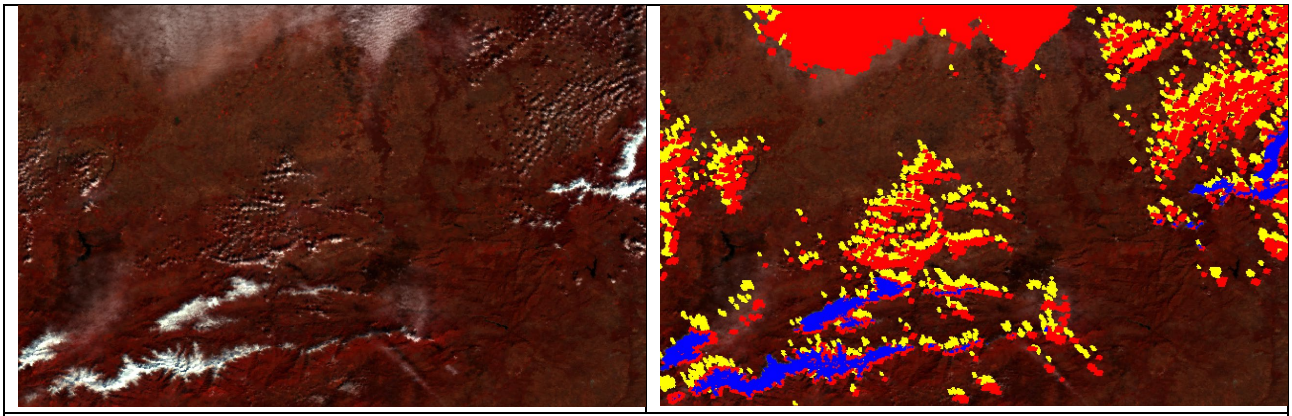

Figure 60: Product cgl_TOC_20181101105411_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good performance of snow/ice detection, although with some over-detection of clouds at the borders of snowy areas. Some semi-transparent clouds or cloud edges are not detected.

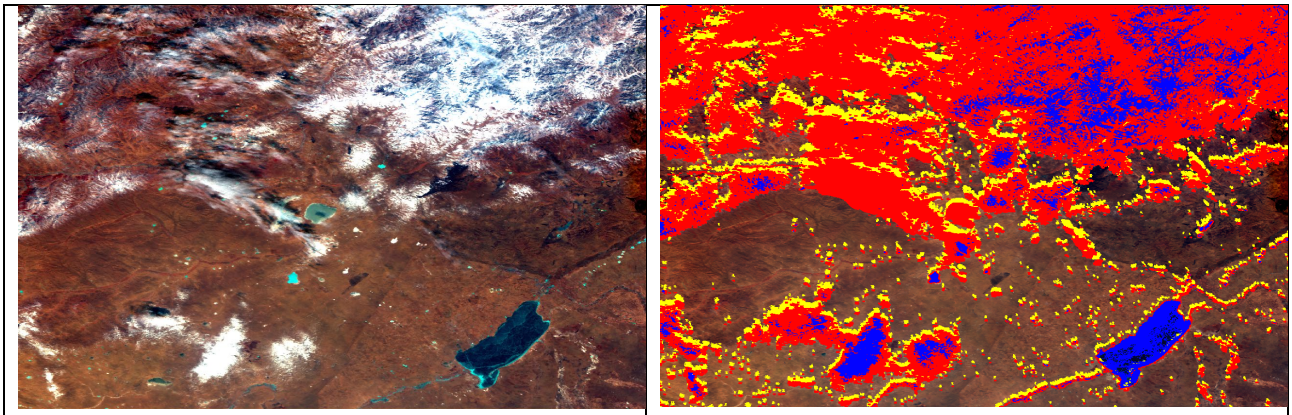

Figure 61: Product cgl_TOC_20190301021759_X29Y03_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows under-detection of snow/ice, large over-detection of clouds, and related cloud shadows over inland water bodies and shorelines.

© CGLOPS Lot-1 consortium

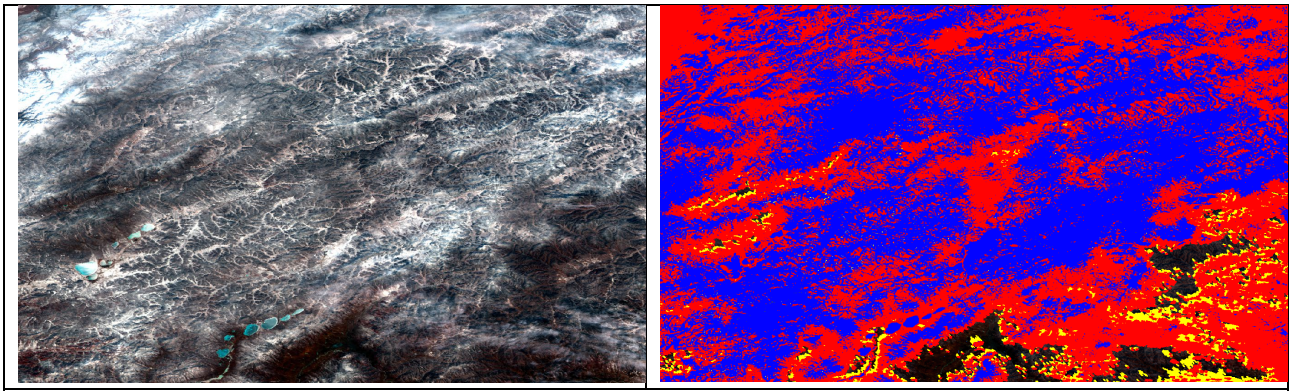

Figure 62: Product cgl_TOC_20190301031630_X29Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows some over-detection of clouds (and associated cloud shadows) over clear sky snow/ice pixels.

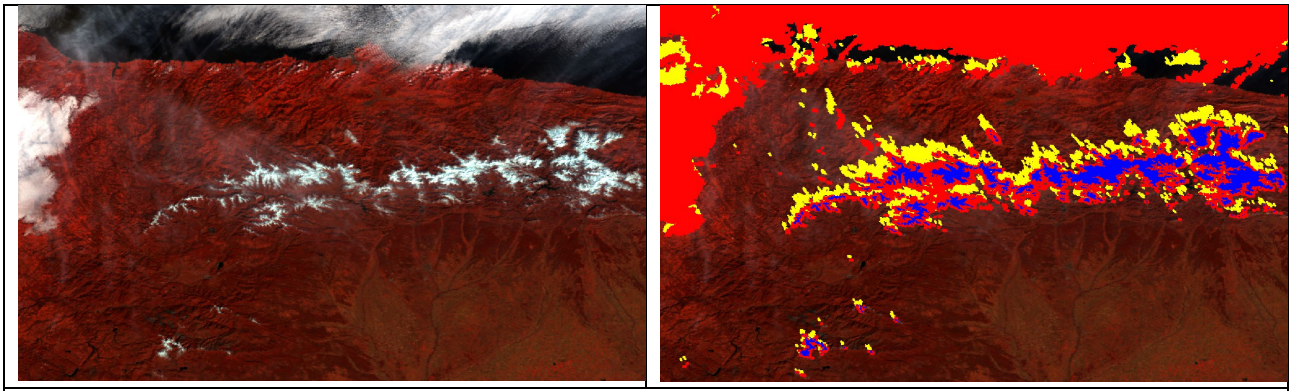

Figure 63: Product cgl_TOC_20190301104255_X17Y04_S3A_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows some over-detection of clouds and associated shadows over snow/ice pixels. There is some under-detection of semi-transparent clouds.

© CGLOPS Lot-1 consortium

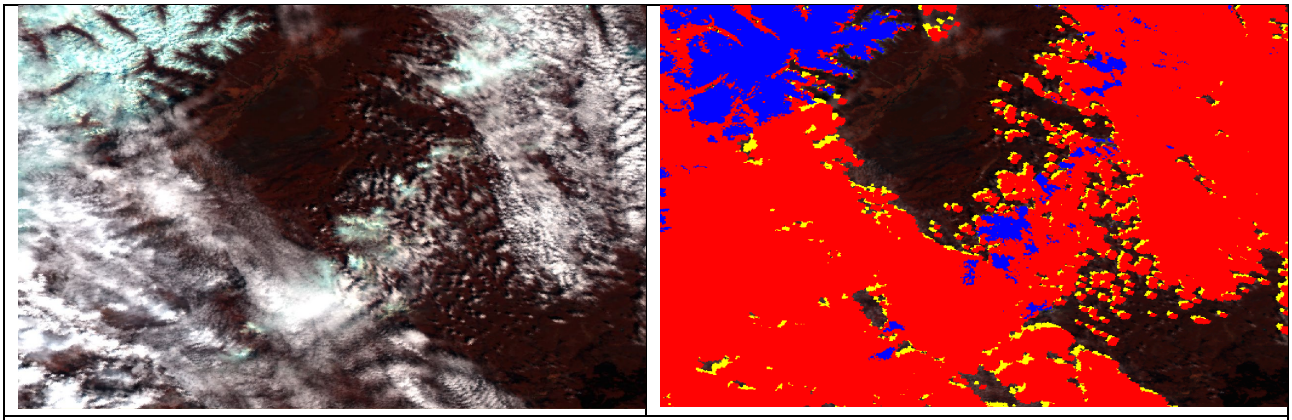

Figure 64: Product cgl_TOC_20190601033126_X29Y03_S3B_v2.3.4 (RGB: Oa17, Oa06, Oa03)

This example shows good distinction between clouds and snow. Some cloud edges and semi-transparent clouds are not detected.

© CGLOPS Lot-1 consortium

4.2 EVALUATION FOR BURNED AREAS DETECTION

The below figures show the comparison between the Sentinel-3 images and the quality flags relevant to the BA algorithm. The Sentinel-3 images are shown as RGB composites, using the SLSTR S6 band (centered at wavelength 2.26 µm) on the red channel, and the OLCI Oa17 and Oa08 bands (centered at 0.865 and 0.665 µm respectively) on the green and blue channels. On each figure, there is a separate panel for each flag (cloud_an for SLSTR and pixel_classif_flag for OLCI), and pixels flagged by a quality indicator are highlighted in yellow. For each example, the Product ID is mentioned, according to the naming standard specified in Section 3.1.

These observations tell us, in relation to accurately mapping burned area, that:

- The IdePix cloud flags, on the 'pixel_classif_flags', are relatively accurate.
- The IdePix cloud shadow flag tends to underestimate cloud shadows.
- The IdePix snow flag tends to pick up areas of light snow although some snow-covered areas are missed.
- There seems to be an over-detection of thin cirrus cloud in the SLSTR 'cloud_an' layer. No other issues have been detected in the SLSTR layer that warrant concern especially if used alongside the IdePix quality flag layers.
- The absence of a cloud shadow layer for SLSTR is not a critical issue for burned area mapping as IdePix includes a cloud shadow layer.

© CGLOPS Lot-1 consortium

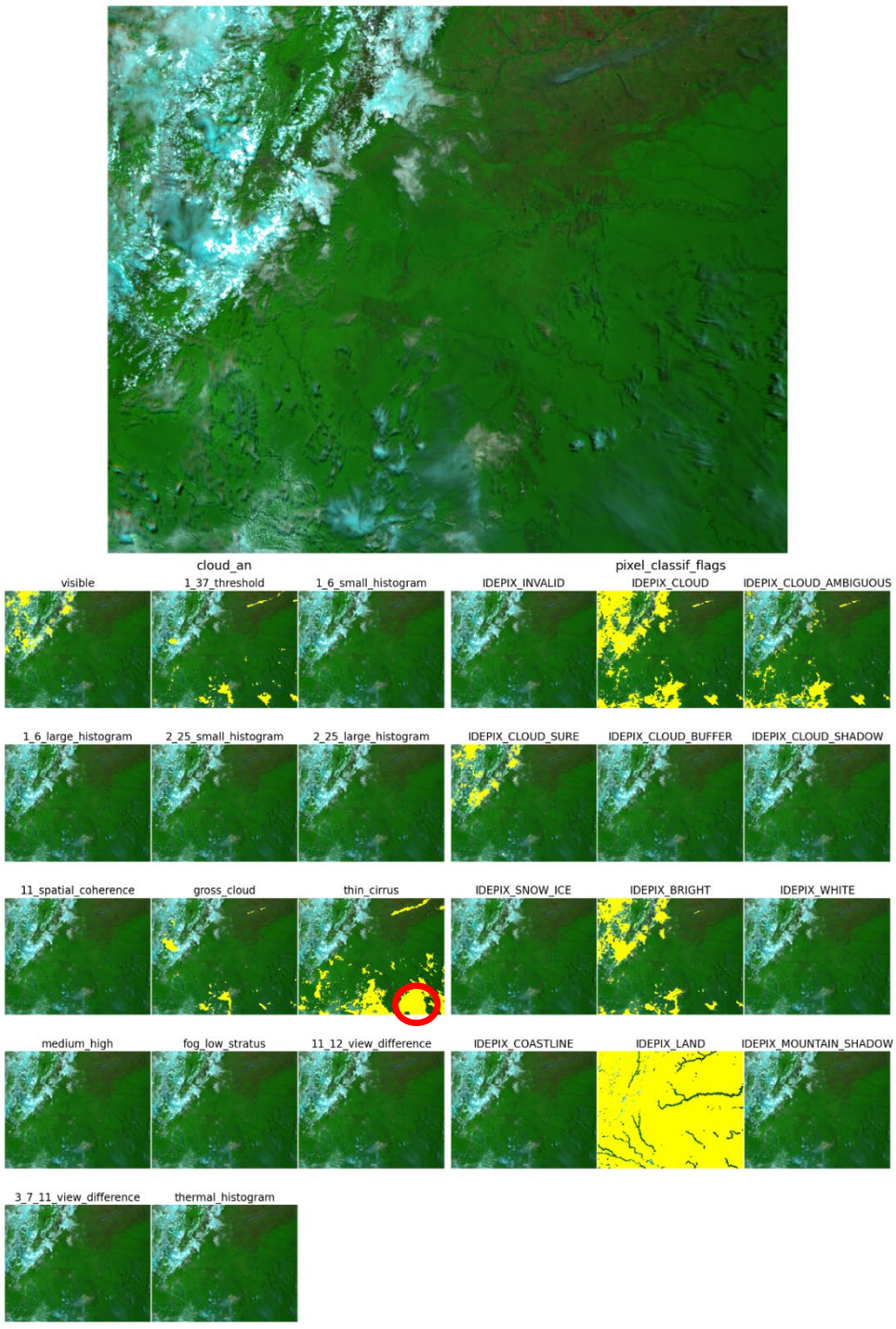

Figure 65: Product cgl_TOC_20190404143538_X10Y08_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows large over-estimation of cloud by the "thin_cirrus" from the 'cloud_an' layer (red circle).

© CGLOPS Lot-1 consortium

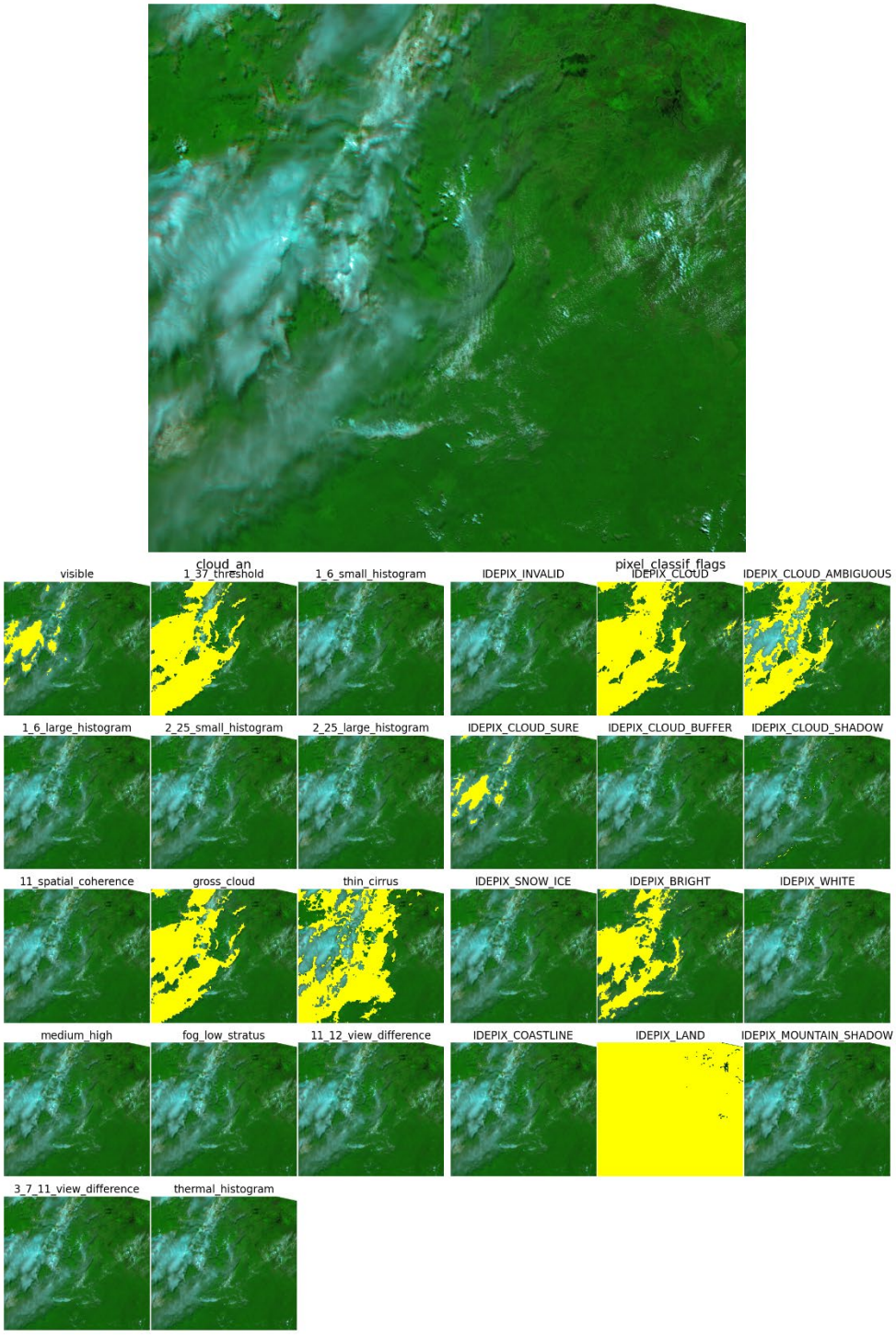

Figure 66: Product cgl_TOC_20180804075137_X20Y07_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows an under-detection of clouds (center-right of the image) by both quality layers. Cloud shadows are mostly missed.

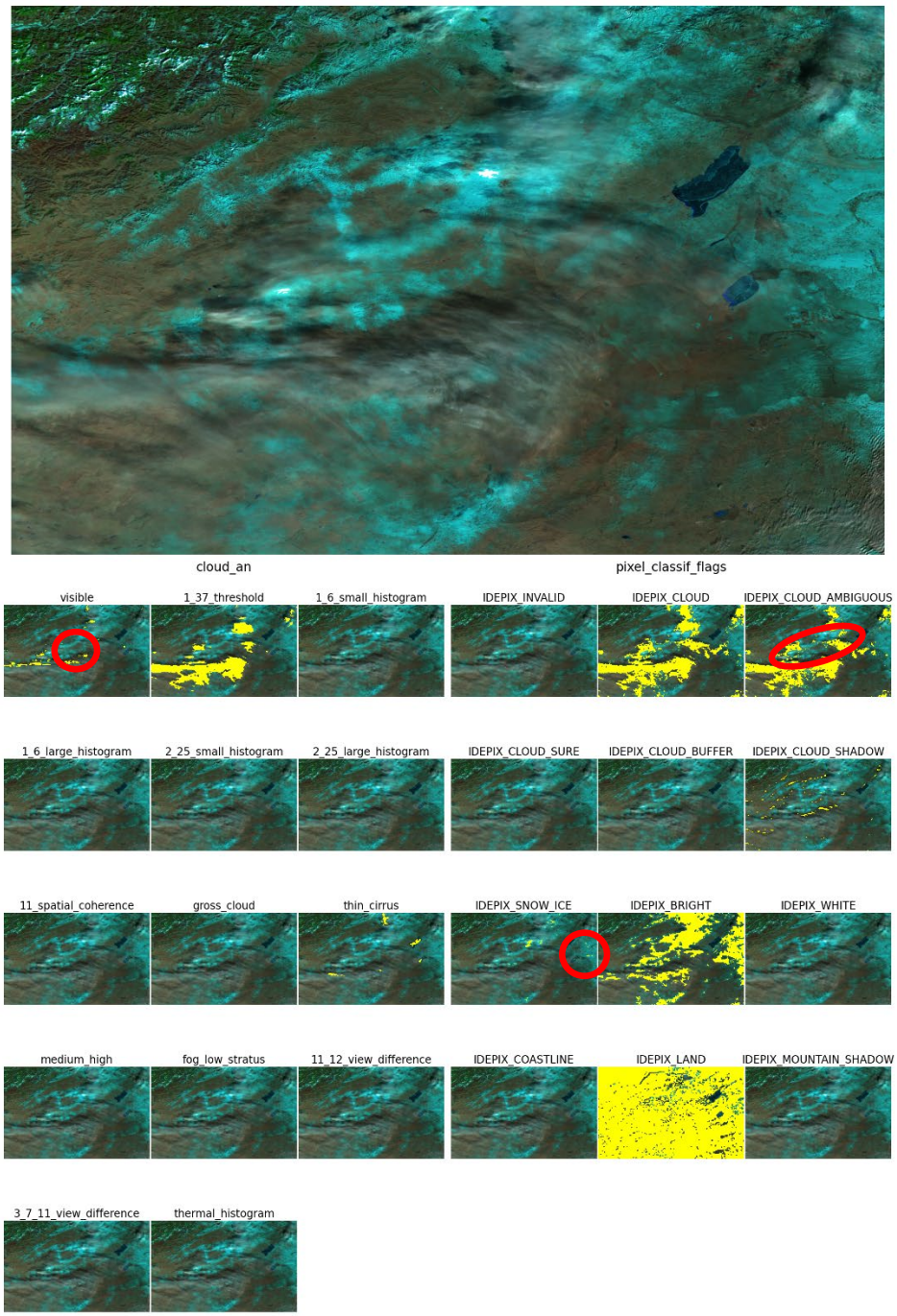

Figure 67: Product cgl_TOC_20190113023640_X29Y03_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows good detection of clouds on the 'pixel_classif_flags' layer, although with quite large under-estimations of cloud shadow at the center of the image and underestimations of snow (snow cover observed across nearby dates). The 'cloud_an' layer omits some of the clouds.

© CGLOPS Lot-1 consortium

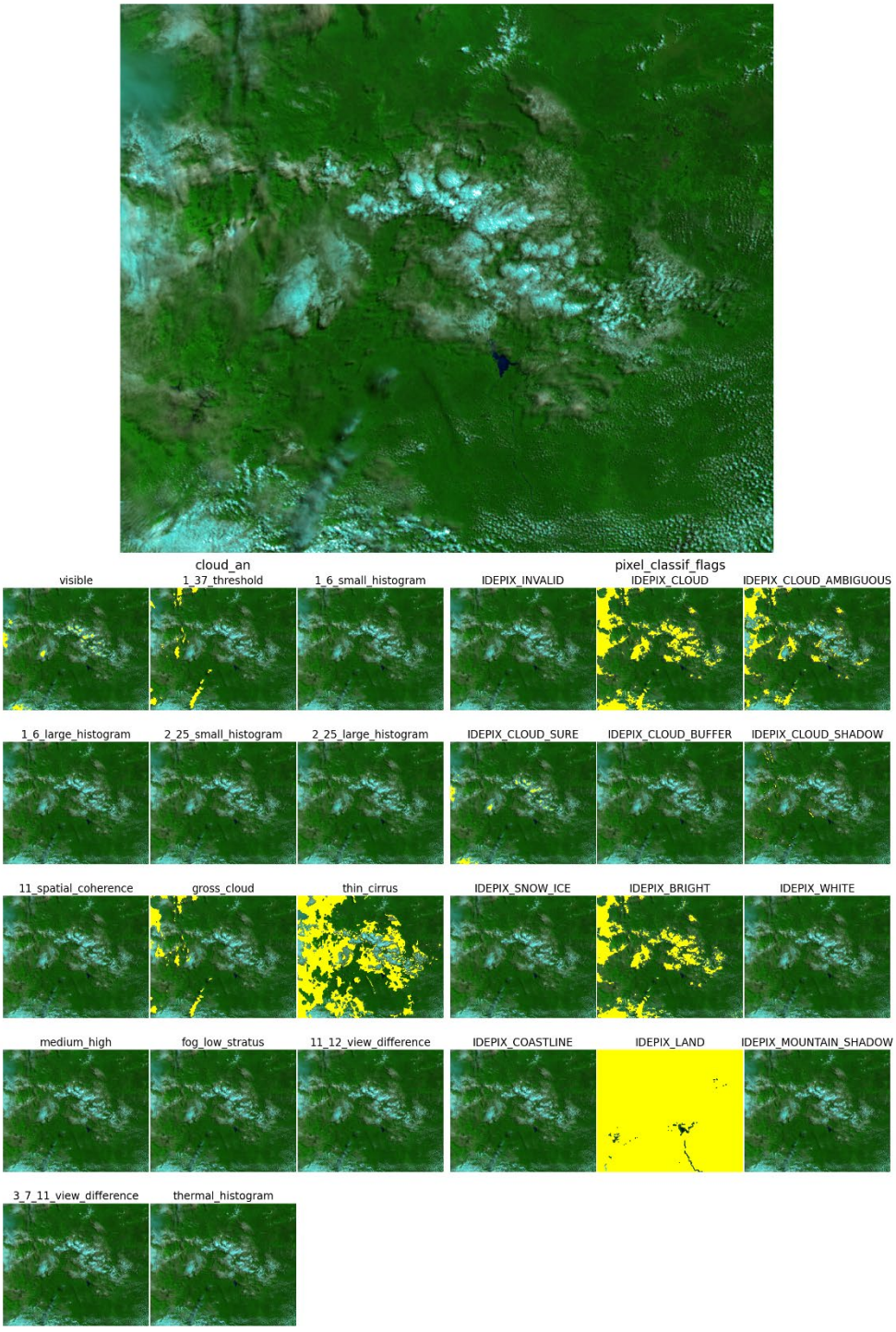

Figure 68: Product cgl_TOC_20180705091013_X19Y07_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows how cloud shadows are mostly missed by the 'pixel_classif_flags'. Some clouds are not detected by any flag, particularly in 'cloud_an'.

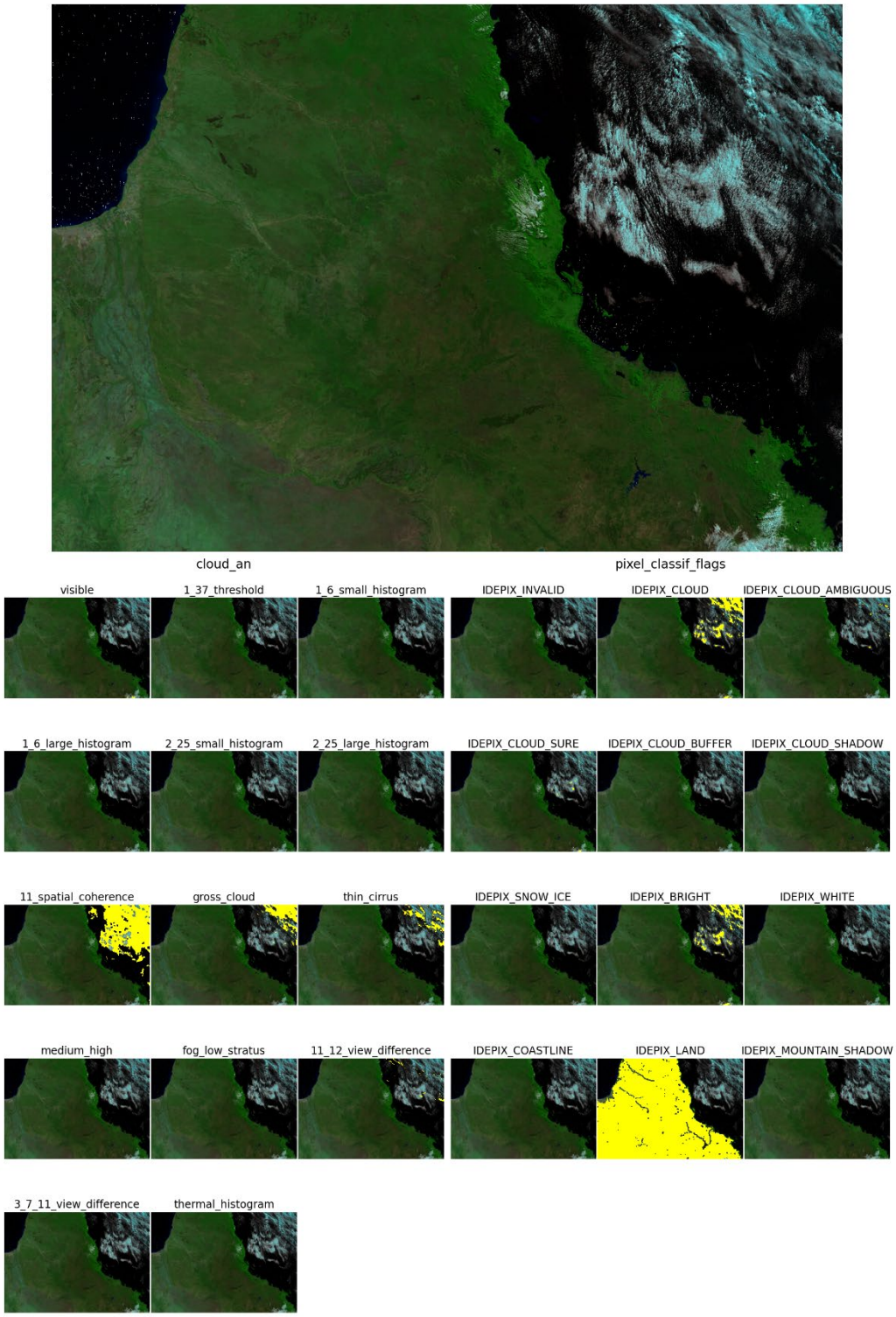

Figure 69: Product cgl_TOC_20190622235859_X32Y10_S3A_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows how some clouds are not detected by any flag, particularly in 'cloud_an'.

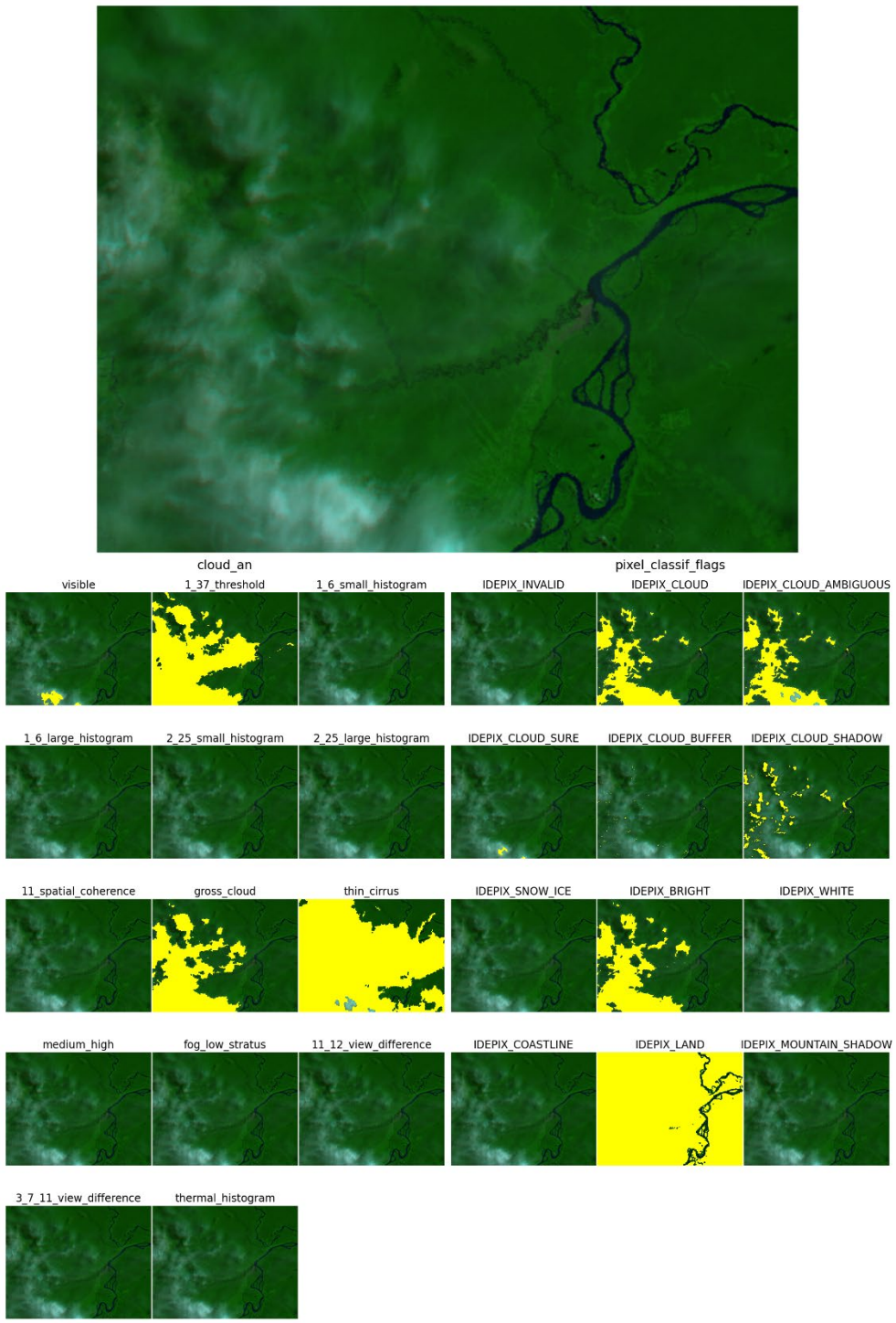

Figure 70: Product cgl_TOC_20190421145902_X10Y08_S3B_v2.3.4.nc (RGB: S6, Oa17, Oa06). This example shows large over-detection of thin cirrus on the 'cloud_an' layer (center-left part of the image).

© CGLOPS Lot-1 consortium

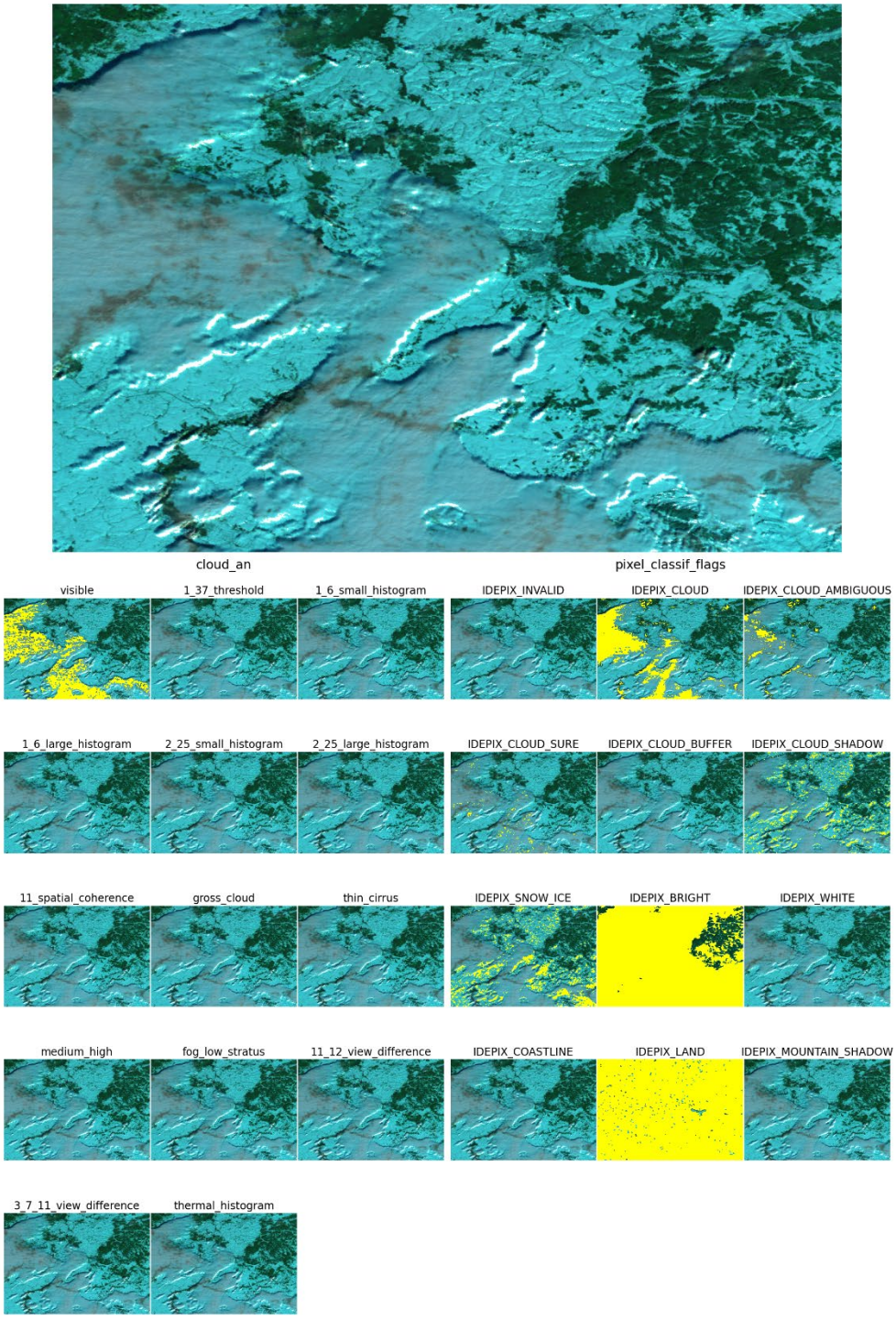

Figure 71: Product cgl_TOC_20190107075436_X22Y03_S3B_v2.3.4.nc (RGB: S1, Oa17, Oa06). This example shows a large under-detection of snow (snow is consistently observed across nearby dates).

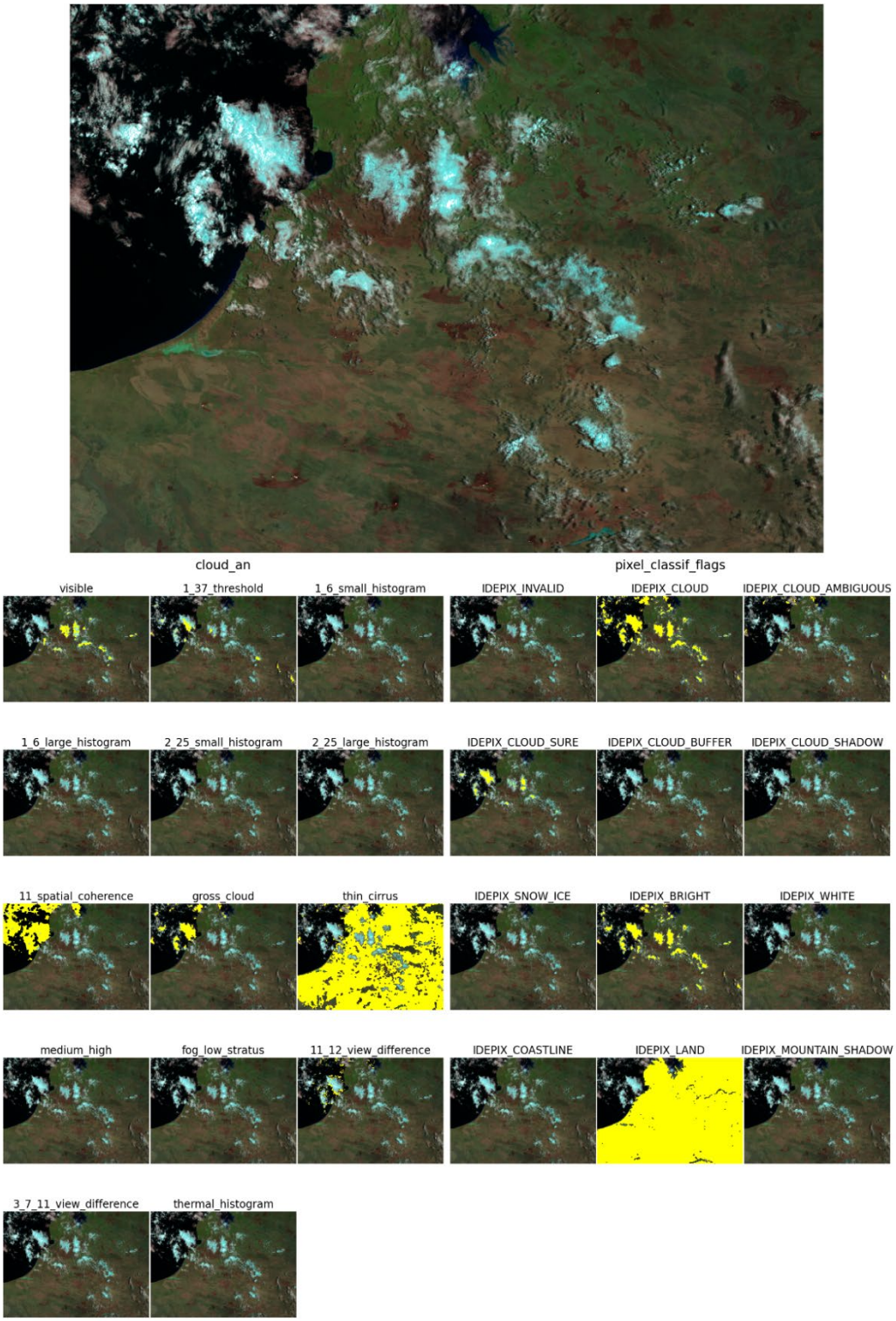

Figure 72: Product cgl_TOC_20181104012842_X30Y10_S3A_v2.3.4.nc (RGB: S1, Oa17, Oa06). This is an example of clearly excessive thin cirrus detection by the 'cloud an' layer.

5 CONCLUSIONS

The evaluation of cloud, cloud shadow and snow masking for the retrieval of vegetation variables and for burned areas detection leads to the following conclusions on IdePix version 9.0.6 cloud, cloud shadow and snow masking:

- Based on this limited qualitative evaluation, we can conclude that the IdePix cloud mask generally performs well.
- IdePix, however, tends still to omit thin or semi-transparent clouds and cloud edges even if this has been improved in this latest IdePix version.
- In some cases, IdePix over-detects clouds over bright surfaces, such as bare areas. This seems to have worsened in this latest IdePix version. At the opposite, the over-detection of clouds over water shorelines is less visible.
- The IdePix cloud shadow masking generally performs well.
- The snow mask of IdePix has improved in this latest IdePix version. More clear sky snow pixels are correctly identified as snow but, in most cases, also the cloud flag is raised. Also, in some cases, the snow/ice flag is still raised over deep convective clouds. However, because the cloud mask, in the main, detects snow cover and vice versa, all snow and cloud pixels seem to the masked. For most of the downstream CGLS vegetation variables, this makes no difference: pixels contaminated by clouds, cloud shadow or snow are excluded from LAI, FAPAR, DMP, NPP and FCOVER processing. Prolonged periods of missing vegetation variables in winter can thus occur, due to snow presence and/or bad illumination conditions. The correct identification of snow/ice (without raising the cloud flag) is relevant for algorithms that require adequate snow detection (e.g. BRDF correction for surface albedo products).

Regarding the SLSTR cloud flag (‘cloud_an’), this layer performs well to mask clouds but there is over-detection of the thin cirrus mask. The thin cirrus mask is therefore not used on the BA algorithm, as it severely limits the number of observations of the land surface. Since the ‘cloud_an’ layer does not contain cloud shadow identification, IdePix is needed to detect cloud shadows in SLSTR. No residual cloud shadows in the SLSTR S5 data channel were observed meaning that the IdePix cloud shadow mask can be utilised for SLSTR channel masking if required.

© CGLOPS Lot-1 consortium

6 REFERENCES

Wevers, J., Müller, D., Danne, O., Brockmann, C., Paperin, M., Stelzer, K., Preusker, R., 2019. Sentinel-3 OLCI Level 2 cloud flagging status and potential improvements, in: 5th S3 Validation Team Meeting. https://nikal.eventsair.com/QuickEventWebsitePortal/sentinel-3-validation-team-s3vt/esa/ExtraContent/ContentSubPage?page=10&subPage=1.

© CGLOPS Lot-1 consortium